

Supporting Information

2-Tridecyn-1-ol 6a: The product was obtained as a pale yellow solid in 3.33 g (85% yield), m.p. 29-31°C (pentane). ^1H NMR (400 MHz, CDCl_3): δ 4.25 (broad s, 2H); 2.21 (*tt*, $J = 2.2/7.1$, 2H); 1.61 (broad s, 1H); 1.45-1.55 (*m*, 2H); 1.20-1.40 (*m*, 14H); 0.88 (*t*, $J = 7.1$, 3H). ^{13}C NMR (100 MHz, CDCl_3): δ 86.66 (*s*), 78.26 (*s*), 51.42 (*t*), 31.88 (*t*), 29.56 (*t*), 29.50 (*t*), 29.30 (*t*), 29.12 (*t*), 28.86 (*t*), 28.60 (*t*), 22.66 (*t*), 18.72 (*t*), 14.08 (*q*). IR (cm^{-1}): 3608. MS (m/z): 169 (1).

2-Pentadecyn-1-ol 6c: The product was obtained as white crystals in 3.44 g (77% yield), m.p. 43-45°C (pentane). ^1H NMR (400 MHz, CDCl_3): δ 4.25 (*dt*, $J = 2.2/6.2$, 2H); 2.21 (*tt*, $J = 2.2/7.1$, 2H); 1.47-1.55 (*m*, 2H); 1.20-1.45 (*m*, 19H); 0.88 (*t*, $J = 7.0$, 3H). ^{13}C NMR (50 MHz, CDCl_3): δ 86.71 (*s*), 78.22 (*s*), 51.46 (*t*), 31.91 (*t*), 29.66 (*t*), 29.63 (*t*), 29.52 (*t*), 29.35 (*t*), 29.14 (*t*), 28.88 (*t*), 28.60 (*t*), 22.68 (*t*), 18.72 (*t*), 14.12 (*q*). IR (cm^{-1}): 3610. MS (m/z): 206 (0.3).

2-Octadecyn-1-ol 6d: The product was obtained as white crystals in 4.18 g (79% yield), m.p. 62-64°C (pentane). ^1H NMR (400 MHz, CDCl_3): δ 4.25 (*dt*, $J = 0.22/6.2$, 2H); 2.21 (*tt*, $J = 2.2/7.1$, 2H); 1.45-1.58 (*m*, 3H); 1.20-1.41 (*m*, 24H); 0.88 (*t*, $J = 7.1$, 3H). ^{13}C NMR (100 MHz, CDCl_3): δ 86.68 (*s*), 78.27 (*s*), 51.43 (*t*), 31.91 (*t*), 29.67 (*t*), 29.64 (*t*), 29.62 (*t*), 29.51 (*t*), 29.34 (*t*), 29.13 (*t*), 28.87 (*t*), 28.61 (*t*), 22.67 (*t*), 18.73 (*t*), 14.08 (*q*). IR (cm^{-1}): 3609. MS (m/z): 248 (0.2).

2-Heneicosyn-1-ol 6e: The product was obtained as an off-white solid in 4.19 g (68% yield), m.p. 69-71°C (pentane). ^1H NMR (400 MHz, CDCl_3): δ 4.26 (broad s, 2H); 2.21 (*tt*, $J = 2.0/7.0$, 2H); 1.45-1.65 (*m*, 3H); 1.15-1.43 (*m*, 30H); 0.89 (*t*, $J = 6.9$, 3H). ^{13}C NMR (100 MHz, CDCl_3): δ 86.68 (*s*), 78.29 (*s*), 51.44 (*t*), 31.91 (*t*), 29.68 (*t*), 29.65 (*t*), 29.62 (*t*), 29.51 (*t*), 29.34 (*t*), 29.13 (*t*), 28.88 (*t*), 28.62 (*t*), 22.67 (*t*), 18.73 (*t*), 14.06 (*q*). IR (cm^{-1}): 3610. MS (m/z): 290 (2).

12-Tridecyn-1-ol 7a: The product was obtained as white crystalline flakes in 1.38 g (70% yield), m.p. 30-33°C (pentane). ^1H NMR (400 MHz, CDCl_3): δ 3.63 (*t*, $J = 6.6$, 2H); 2.18 (*td*, $J = 2.7/7.1$, 2H); 1.94 (*t*, $J = 2.64$, 1H); 1.47-1.60 (*m*, 4H); 1.20-1.45 (*m*, 15H). ^{13}C NMR (100 MHz, CDCl_3): δ 84.74 (*s*), 67.99 (*d*), 63.00 (*t*), 32.75 (*t*), 29.51 (*t*), 29.45 (*t*), 29.41 (*t*), 29.36 (*t*), 29.03 (*t*), 28.69 (*t*), 28.44 (*t*), 25.69 (*t*), 18.34 (*t*). IR (cm^{-1}): 3622, 3307. MS (m/z): 149 (1). Analysis for $\text{C}_{13}\text{H}_{24}\text{O}$: C = 79.40%; H = 12.32% (calcd. C = 79.52%; H = 12.33%).

14-Pentadecyn-1-ol 7c: The product was obtained as white crystalline flakes in 2.38 g (76% yield), m.p. 45-48°C (pentane). ^1H NMR (400 MHz, CDCl_3): δ 3.62 (*t*, $J = 6.6$, 2H); 2.17 (*td*, $J = 2.6/7.0$, 2H); 1.92 (*t*, $J = 2.6$, 1H); 1.28-1.56 (*m*, 4H); 1.25-1.40 (*m*, 19H). ^{13}C NMR (50 MHz, CDCl_3): δ 84.79 (*s*), 68.01 (*d*), 63.06 (*t*), 32.79 (*t*), 29.56 (*t*), 29.48 (*t*), 29.41 (*t*), 29.09 (*t*), 28.75 (*t*), 28.47 (*t*), 25.72 (*t*), 18.38 (*t*). IR (cm^{-1}): 3624, 3307. MS (m/z): 177 (0.3). Analysis for $\text{C}_{15}\text{H}_{28}\text{O}$: C = 80.03%; H = 12.84% (calcd. C = 80.28%; H = 12.59%).

17-Octadecyn-1-ol 7d: The product was obtained as white crystalline flakes in 3.26 g (82% yield), m.p. 62-64°C (pentane). ^1H NMR (400 MHz, CDCl_3): δ 3.63 (q), and on addition of D_2O (t , $J = 6.6$, 2H); 2.18 (td , $J = 2.6/7.0$, 2H); 1.94 (t , $J = 2.6$, 1H); 1.47-1.60 (m , 4H); 1.20-1.45 (m , 25H). ^{13}C NMR (100 MHz, CDCl_3): δ 84.80 (s), 67.99 (d), 63.09 (t), 32.82 (t), 29.64 (t), 29.59 (t), 29.49 (t), 29.42 (t), 28.76 (t), 28.50 (t), 25.73 (t), 18.39 (t). IR (cm^{-1}): 3623, 3307. MS (m/z): 267 (0.02). Analysis for $\text{C}_{18}\text{H}_{34}\text{O}$: C = 80.96%; H = 12.90% (calcd. C = 81.12%; H = 12.87%).

20-Heneicosyn-1-ol 7e: The product was obtained as an off-white crystalline solid in 1.18 g (77% yield), m.p. 71-73°C (pentane). ^1H NMR (400 MHz, CDCl_3): δ 3.65 (t , $J = 6.6$, 2H); 2.19 (td , $J = 2.7/7.1$, 2H); 1.94 (t , $J = 2.7$, 1H); 1.48-1.61 (m , 4H); 1.24-1.45 (m , 31H). ^{13}C NMR (100 MHz, CDCl_3): δ 84.77 (s), 67.97 (d), 63.10 (t), 32.81 (t), 29.64 (t), 29.57 (t), 29.47 (t), 29.41 (t), 29.08 (t), 28.74 (t), 28.49 (t), 25.72 (t), 18.37 (t). IR (cm^{-1}): 3622, 3307. MS (m/z): 308 (0.2). Analysis for $\text{C}_{21}\text{H}_{40}\text{O}$: C = 81.53%; H = 13.16% (calcd. C = 81.74%; H = 13.08%).

13-Tetradecyn-1-ol 7b: To a stirred solution of **10b** (8.0 g, 18 mmol) in dry ether (40 mL) were added 4-toluenesulfonic acid monohydrate (130 mg, 0.6 mmol) followed by a dropwise addition of 3,4-dihydro-2-*H*-pyran (4.5 mL, 50 mmol) at room temperature. The reaction mixture was stirred at room temperature overnight, then quenched with 6 N aq. NaOH solution (100 mL). The organic phase was separated, dried (MgSO_4), and concentrated. The residue was dissolved in dry degassed DMSO (30 mL) and treated at 5°C with a dropwise addition of a suspension of lithium acetylide ethylenediamine complex (3.5 g, 38.0 mmol) in DMSO (30 mL).

After complete addition, the reaction mixture was stirred overnight at room temperature, then poured into ice-water and extracted with ether. The combined ether extracts were washed with water, dried (MgSO_4), concentrated and the remaining yellow oily residue dissolved in methanol (50 mL) and 4-toluenesulfonic acid monohydrate (1.0 g) was added at room temperature. The reaction mixture was stirred at room temperature overnight, then poured into ice-water and extracted with ether. The combined ether extracts were washed with water, dried (MgSO_4), concentrated and flash chromatographed (SiO_2 ; *n*-hexane : ether = 2/1). The product was obtained as white crystals in 4.5 g (72% yield), m.p. 40-42°C (pentane). ^1H NMR (400 MHz, CDCl_3): δ 3.64 (*t*, J = 6.6, 2H); ; 2.18 (*td*, J = 2.6/7.0, 2H); 1.94 (*t*, J = 2.6, 1H); 1.48-1.61 (*m*, 4H); 1.24-1.44 (*m*, 17H). ^{13}C NMR (100 MHz, CDCl_3): δ 84.77 (*s*), 68.01 (*d*), 63.02 (*t*), 32.78 (*t*), 29.57 (*t*), 29.56 (*t*), 29.54 (*t*), 29.46 (*t*), 29.40 (*t*), 29.07 (*t*), 28.73 (*t*), 28.47 (*t*), 25.72 (*t*), 18.37 (*t*). IR (cm^{-1}): 3625, 3307. MS (*m/z*): 149 (1). Analysis for $\text{C}_{14}\text{H}_{26}\text{O}$: C = 79.86%; H = 12.61% (calcd. C = 79.92%; H = 12.47%).

12-Tridecynal 8a: The product was obtained as a colourless oil in 1.18 g (76% yield). ^1H NMR (400 MHz, CDCl_3): δ 9.77 (*t*, J = 1.8, 1H); ; 2.42 (*td*, J = 1.8/7.5, 2H); 2.18 (*td*, J = 2.6/7.1, 2H); 1.94 (*t*, J = 2.6, 1H); 1.48-1.70 (*m*, 4H); 1.20-1.45 (*m*, 12H). ^{13}C NMR (100 MHz, CDCl_3): δ 202.83 (*d*), 84.72 (*s*), 68.02 (*d*), 43.87 (*t*), 29.35 (*t*), 29.30 (*t*), 29.28 (*t*), 29.11 (*t*), 29.01 (*t*), 28.68 (*t*), 28.43 (*t*), 22.04 (*t*), 18.35 (*t*). IR (cm^{-1}): 3307, 1721. MS (*m/z*): 161 (0.4).

13-Tetradecynal 8b: The product was obtained as a colourless oil in 1.45 g (87% yield). ^1H NMR (400 MHz, CDCl_3): δ 9.77 (*t*, J = 1.8, 1H); 2.42 (*td*, J = 1.8/7.1,

2H); 2.18 (*td*, $J = 2.6/7.1$, 2H); 1.94 (*t*, $J = 2.6$, 1H); 1.49-1.70 (*m*, 4H); 1.28-1.40 (*m*, 14H). ^{13}C NMR (50 MHz, CDCl_3): δ 202.89 (*d*), 84.76 (*s*), 68.02 (*d*), 43.89 (*t*), 29.47 (*t*), 29.43 (*t*), 29.36 (*t*), 29.31 (*t*), 29.13 (*t*), 29.06 (*t*), 28.72 (*t*), 28.46 (*t*), 22.05 (*t*), 18.37 (*t*). IR (cm^{-1}): 3307, 1721. MS (m/z): 149 (1).

14-Pentadecynal 8c: The product was obtained as a white crystalline solid in 1.77 g (81% yield), m.p. 35-36°C (pentane). ^1H NMR (400 MHz, CDCl_3): δ 9.77 (*t*, $J = 1.8$, 1H). 2.42 (*td*, $J = 1.8/7.5$, 2H); 2.18 (*td*, $J = 2.6/7.1$, 2H); 1.94 (*t*, $J = 2.7$, 1H); 1.48-1.70 (*m*, 4H); 1.27-1.40 (*m*, 16H). ^{13}C NMR (50 MHz, CDCl_3): δ 202.93 (*d*), 84.78 (*s*), 68.02 (*d*), 43.90 (*t*), 29.52 (*t*), 29.45 (*t*), 29.39 (*t*), 29.33 (*t*), 29.14 (*t*), 29.08 (*t*), 28.74 (*t*), 28.47 (*t*), 22.06 (*t*), 18.38 (*t*). IR (cm^{-1}): 3307, 1721. MS (m/z): 222 (1).

17-Octadecynal 8d: The product was obtained as white crystals in 0.72 g (52% yield), m.p. 45-47°C (pentane). ^1H NMR (400 MHz, CDCl_3): δ 9.77 (*t*, $J = 1.8$, 1H). 2.42 (*td*, $J = 1.8/7.5$, 2H); 2.18 (*td*, $J = 2.6/7.1$, 2H); 1.94 (*t*, $J = 2.7$, 1H); 1.47-1.70 (*m*, 4H); 1.20-1.45 (*m*, 22H). ^{13}C NMR (100 MHz, CDCl_3): δ 202.92 (*d*), 84.82 (*s*), 68.01 (*d*), 43.92 (*t*), 29.63 (*t*), 29.59 (*t*), 29.57 (*t*), 29.50 (*t*), 29.42 (*t*), 29.35 (*t*), 29.17 (*t*), 29.11 (*t*), 28.50 (*t*), 22.10 (*t*), 18.40 (*t*). IR (cm^{-1}): 3307, 1721. MS (m/z): 264 (0.2).

20-Heneicosynal 8e: The product was obtained as a pale yellow solid in 0.77 g (72% yield), m.p. 59-61°C (pentane). ^1H NMR (400 MHz, CDCl_3): δ 9.76 (*t*, $J = 1.8$, 1H); 2.42 (*td*, $J = 1.8/7.1$, 2H); 2.18 (*td*, $J = 2.6/7.1$, 2H); 1.94 (*t*, $J = 2.6$, 1H); 1.47-1.70 (*m*, 4H); 1.20-1.45 (*m*, 28H). ^{13}C NMR (100 MHz, CDCl_3): δ 202.91 (*d*),

84.81 (*s*), 68.01 (*d*), 43.92 (*t*), 29.67 (*t*), 29.61 (*t*), 29.58 (*t*), 29.50 (*t*), 29.43 (*t*), 29.35 (*t*), 29.17 (*t*), 29.11 (*t*), 28.77 (*t*), 28.51 (*t*), 22.10 (*t*), 18.40 (*t*). IR (cm^{-1}): 3307, 1721. MS (*m/z*): 307 (0.1).

9-Hydroxynonanoic acid methyl ester 11b: NaBH_4 (1.41 g, 36 mmol) was suspended in dry THF (60 mL) with stirring and a solution of monomethyl azelate (5.81 mL, 30 mmol) in dry THF (60 mL) was added dropwise at room temperature during 30 min. The mixture was then stirred 1 h at room temperature, then a solution of iodine (3.81 g, 15 mmol) in dry THF (60 mL) was added slowly over 1 h at 0°C (ice bath cooling). The mixture was stirred additionally for 1 h at 0°C , then 3 N aq. HCl solution (15 mL) was added carefully and the mixture extracted with ether. The combined extracts were washed with 3 N aq. NaOH solution, brine to pH = 7 and dried (MgSO_4). After concentration and flash chromatography (SiO_2 ; *n*-hexane : ether = 1/1) the product was isolated as a yellow oil in 4.27 g (76% yield). ^1H NMR (400 MHz, CDCl_3): δ 3.67 (*s*, 3H); 3.65 (*t*, $J = 6.4$, 2H); 2.31 (*t*, $J = 7.6$, 2H); 1.54-1.64 (*m*, 4H); 1.30-1.44 (*m*, 9H). ^{13}C NMR (100 MHz, CDCl_3): δ 174.29 (*s*), 63.00 (*t*), 51.43 (*q*), 34.07 (*t*), 32.74 (*t*), 29.17 (*t*), 29.03 (*t*), 25.64 (*t*), 24.89 (*t*). IR (cm^{-1}): 3622, 1729. MS (*m/z*) : 189 (21). This compound was found to be of limited stability and was used immediately in the next step.

ω -((1,1-Dimethylethyl)diphenylsilyloxy)carboxylic acid methyl ester 12:

Imidazole (3.4 g, 50 mmol), and **11** (20 mmol) were dissolved in dry DMF (11.5 mL) and (1,1-dimethylethyl)diphenylchlorosilane (6.2 mL, 24.22 mmol) was added dropwise at room temperature. The reaction is slightly exothermic and became turbid

immediately. The reaction mixture was stirred overnight at room temperature, then poured into ice-water and extracted with pentane. After drying (MgSO_4), concentration and flash chromatography (SiO_2 ; *n*-hexane : ether = 15/1) the protected alcohol was isolated in a pure form.

6-((1,1-Dimethylethyl)diphenylsilyloxy)hexanoic acid methyl ester 12a:

The product was obtained as a colourless oil in 8.85 g (92% yield). ^1H NMR (400 MHz, CDCl_3): δ 7.60-7.70 (*m*, 4H); 7.35-7.45 (*m*, 6H); 3.70 (*s*, 3H); 3.65 (*t*, $J = 6.3$, 2H); 2.29 (*t*, $J = 7.6$, 2H); 1.30-1.70 (*m*, 6H); 1.04 (*s*, 9H). ^{13}C NMR (100 MHz, CDCl_3): δ 174.16 (*s*), 135.56 (*d*), 134.08 (*s*), 129.51 (*d*), 127.58 (*d*), 63.67 (*t*), 51.42 (*q*), 34.10 (*t*), 32.17 (*t*), 26.86 (*q*), 25.41 (*t*), 24.74 (*t*), 19.21 (*s*). IR (cm^{-1}): 1728. MS (m/z): 369 (0.3).

9-((1,1-Dimethylethyl)diphenylsilyloxy)nonanoic acid methyl ester 12b:

The product was obtained as a colourless oil in 7.54 g (88% yield). ^1H NMR (400 MHz, CDCl_3): δ 7.60-7.70 (*m*, 4H); 7.35-7.50 (*m*, 6H); 3.66 (*s*, 3H); 3.64 (*t*, $J = 7.2$, 2H); 2.31 (*t*, $J = 7.5$, 2H); 1.52-1.63 (*m*, 4H); 1.26-1.38 (*m*, 8H); 1.06 (*s*, 9H). ^{13}C NMR (100 MHz, CDCl_3): δ 174.29 (*s*), 135.58 (*d*), 134.21 (*s*), 129.47 (*d*), 127.55 (*d*), 63.97 (*t*), 51.41 (*q*), 34.11 (*t*), 32.53 (*t*), 29.20 (*t*), 29.16 (*t*), 29.09 (*t*), 26.88 (*q*), 25.70 (*t*), 24.94 (*t*), 19.22 (*s*). IR (cm^{-1}): 1731. MS: 427 (0.3).

ω -((1,1-Dimethylethyl)diphenylsilyloxy)carboxylic acid 13: To a solution of **12** (22 mmol) in MeOH (105 mL) and water (52 mL) was added K_2CO_3 (8 g, 57.88 mmol) and the mixture refluxed with stirring for 2 h where a clear colourless solution

was formed showing a complete conversion. MeOH was evaporated at a rotary evaporator and the remaining aq. solution saturated with solid NaCl and acidified to pH 1 with 3 N aq. HCl solution. The mixture was extracted with ether and the combined extracts washed with brine to pH = 7, dried (MgSO₄) and concentrated to afford the free carboxylic acid.

6-((1,1-Dimethylethyl)diphenylsilyloxy)hexanoic acid 13a: The product was obtained as a colourless oil in 7.74 g (95% yield). ¹H NMR (400 MHz, CDCl₃): δ 7.65-7.70 (*m*, 4H); 7.35-7.45 (*m*, 6H); 3.67 (*t*, *J* = 6.2, 2H); 2.35 (*t*, *J* = 7.6, 2H); 1.50-1.68 (*m*, 4H); 1.30-1.48 (*m*, 2H); 1.06 (*s*, 9H). ¹³C NMR (50 MHz, CDCl₃): δ 179.39 (*s*), 135.56 (*d*), 134.00 (*s*), 127.59 (*d*), 63.58 (*t*), 33.92 (*t*), 32.12 (*t*), 26.85 (*q*), 25.30 (*t*), 24.42 (*t*), 19.20 (*s*). IR (cm⁻¹): 3048, 1709. MS (*m/z*): 313 (0.7).

9-((1,1-Dimethylethyl)diphenylsilyloxy)nonanoic acid 13b: The product was obtained as a colourless oil in 6.90 g (99% yield). ¹H NMR (400 MHz, CDCl₃): δ 7.64-7.70 (*m*, 4H); 7.35-7.43 (*m*, 6H); 3.65 (*t*, *J* = 6.4, 2H); 2.34 (*t*, *J* = 7.5, 2H); 1.51-1.64 (*m*, 4H); 1.27-1.35 (*m*, 8H); 1.04 (*s*, 9H). ¹³C NMR (100 MHz, CDCl₃): δ 179.06 (*s*), 135.58 (*d*), 134.20 (*s*), 129.47 (*d*), 127.55 (*d*), 63.96 (*t*), 33.87 (*t*), 32.52 (*t*), 29.18 (*t*), 29.13 (*t*), 28.99 (*t*), 26.88 (*q*), 25.69 (*t*), 24.67 (*t*), 19.22 (*s*). IR (cm⁻¹): 2500-3500 (broad band), 1708. MS (*m/z*): 413 (0.3).

6-((1,1-Dimethylethyl)diphenylsilyloxy)hexanoic acid 5-hexynyl ester 14a: The product was obtained as a colourless oil in 2.06 g (65% yield). ¹H NMR (400 MHz, CDCl₃): δ 7.60-7.70 (*m*, 4H); 7.32-7.45 (*m*, 6H); 4.08 (*t*, *J* = 6.6, 2H); 3.65 (*t*, *J*

= 6.4, 2H); 2.28 (*t*, *J* = 7.5, 2H); 2.22 (*td*, *J* = 2.6/7.1, 2H); 1.95 (*t*, *J* = 2.7, 1H); 1.68-1.85 (*m*, 2H); 1.51-1.65 (*m*, 6H); 1.30-1.48 (*m*, 2H); 1.04 (*s*, 9H). ¹³C NMR (100 MHz, CDCl₃): δ 173.75 (*s*), 135.57 (*d*), 134.10 (*s*), 129.52 (*d*), 127.59 (*d*), 83.86 (*s*), 68.69 (*d*), 63.70 (*t*), 34.34 (*t*), 32.21 (*t*), 27.72 (*t*), 26.88 (*q*) 25.45 (*t*), 24.96 (*t*), 24.78 (*t*), 19.22 (*s*), 18.08 (*t*). IR (cm⁻¹): 3304, 1735. MS (*m/z*): 396 (0.8).

9-((1,1-Dimethylethyl)diphenylsilyloxy)nonanoic acid 5-hexynyl ester 14b:

The product was obtained as a colourless oil in 4.93 g (67% yield). ¹H NMR (400 MHz, CDCl₃): δ 7.65-7.70 (*m*, 4H); 7.35-7.45 (*m*, 6H); 4.10 (*t*, *J* = 6.6, 2H); 3.66 (*t*, *J* = 6.6, 2H); 2.30 (*t*, *J* = 7.5, 2H); 2.24 (*td*, *J* = 2.7/7.1, 2H); 1.96 (*t*, *J* = 2.6, 1H); 1.70-1.80 (*m*, 2H); 1.52-1.65 (*m*, 6H); 1.20-1.38 (*m*, 8H); 1.06 (*s*, 9H). ¹³C NMR (100 MHz, CDCl₃): δ 173.86 (*s*), 135.58 (*d*), 134.22 (*s*), 129.47 (*d*), 127.56 (*d*), 83.86 (*s*), 68.67 (*d*), 63.98 (*t*), 63.67 (*t*), 34.35 (*t*), 32.54 (*t*), 29.21 (*t*), 29.18 (*t*), 29.11 (*t*), 27.73 (*t*), 26.88 (*q*), 25.72 (*t*), 24.97 (*t*), 19.22 (*s*), 18.08 (*t*). IR (cm⁻¹): 3307, 1726. MS (*m/z*): 455 (1).

6-((1,1-Dimethylethyl)diphenylsilyloxy)hexanoic acid 9-decynyl ester 14c:

The product was obtained as a colourless oil in 2.01 g (65% yield). ¹H NMR (400 MHz, CDCl₃): δ 7.62-7.70 (*m*, 4H); 7.35-7.43 (*m*, 6H); 4.06 (*t*, *J* = 6.6, 2H); 3.66 (*t*, *J* = 6.4, 2H); 2.29 (*t*, *J* = 7.5, 2H); 2.18 (*td*, *J* = 2.6/7.1, 2H); 1.95 (*t*, *J* = 2.7, 1H); 1.48-1.65 (*m*, 8H);); 1.25-1.45 (*m*, 10H); 1.05 (*s*, 9H). ¹³C NMR (100 MHz, CDCl₃): δ 173.81 (*s*), 135.56 (*d*), 134.09 (*s*), 129.50 (*d*), 127.58 (*d*), 84.67 (*s*), 68.09 (*d*), 64.35 (*t*), 63.71 (*t*), 34.38 (*t*), 32.20 (*t*), 29.08 (*t*), 28.95 (*t*), 28.62 (*t*), 28.42 (*t*), 26.87 (*q*), 25.86 (*t*), 25.44 (*t*), 24.80 (*t*), 19.20 (*s*), 18.36 (*t*). IR (cm⁻¹): 3308, 1735. MS (*m/z*):

452 (0.7).

6-((1,1-Dimethylethyl)diphenylsilyloxy)hexanoic acid 13-tetradecynyl ester 14d: The product was obtained as a colourless oil in 2.75 g (70% yield). ^1H NMR (400 MHz, CDCl_3): δ 7.63-7.70 (*m* 4H); 7.35-7.45 (*m*, 6H); 4.06 (*t*, $J = 7.1$, 2H); 3.66 (*t*, $J = 6.6$, 2H); 2.29 (*t*, $J = 7.5$, 2H); 2.18 (*td*, $J = 2.6/7.1$, 2H); 1.95 (*t*, $J = 2.6$, 1H); 1.48-1.65 (*m*, 8H); 1.25-1.45 (*m*, 18H); 1.05 (*s*, 9H). ^{13}C NMR (100 MHz, CDCl_3): δ 173.83 (*s*), 135.56 (*d*), 134.10 (*s*), 129.51 (*d*), 127.58 (*d*), 84.79 (*s*), 68.02 (*d*), 64.42 (*t*), 63.71 (*t*), 34.39 (*t*), 34.39 (*t*), 32.21 (*t*), 29.54 (*t*), 29.53 (*t*), 29.48 (*t*), 29.24 (*t*), 29.10 (*t*), 28.75 (*t*), 28.66 (*t*), 26.87 (*q*), 25.45 (*t*), 24.81 (*t*), 19.21 (*s*), 18.40 (*t*). IR (cm^{-1}): 3310, 1736. MS (m/z): 508 (2).

6-Hydroxyhexanoic acid 5-hexynyl ester 15a: The product was obtained as a colourless oil in 0.79 g (93% yield). ^1H NMR (400 MHz, CDCl_3): δ 4.10 (*t*, $J = 6.4$, 2H); 3.66 (*m*, 2H); 2.33 (*t*, $J = 7.6$, 2H); 2.24 (*td*, $J = 2.7/7.0$, 2H); 1.97 (*t*, $J = 2.6$, 1H); 1.55-1.80 (*m*, 9H); 1.30-1.45 (*m*, 2H). ^{13}C NMR (100 MHz, CDCl_3): δ 173.71 (*s*), 83.87 (*s*), 68.70 (*d*), 63.78 (*t*), 62.68 (*t*), 34.22 (*t*), 32.33 (*t*), 27.72 (*t*), 25.30 (*t*), 24.97 (*t*), 24.66 (*t*), 18.08 (*t*). IR (cm^{-1}): 3627, 3307, 1726. MS (m/z): 213 (10). HR-MS: 213.1496 (calcd. for $\text{C}_{12}\text{H}_{21}\text{O}_3$: 213.1491). Analysis for $\text{C}_{12}\text{H}_{20}\text{O}_3$: C = 67.60%; H = 9.55% (calcd. C = 67.88%; H = 9.50%).

9-Hydroxynonanoic acid 5-hexynyl ester 15b: The product was obtained as a colourless oil in 1.85 g (91% yield). ^1H NMR (400 MHz, CDCl_3): δ 4.10 (*t*, $J = 6.4$, 2H); 3.64 (*m*, 2H); 2.30 (*t*, $J = 7.5$, 2H); 2.24 (*td*, $J = 2.7/7.1$, 2H); 1.97 (*t*, $J = 2.4$,

1H); 1.70-1.80 (*m*, 2H); 1.50-1.69 (*m*, 6H); 1.20-1.40 (*m*, 9H). ^{13}C NMR (100 MHz, CDCl_3): δ 173.85 (*s*), 83.85 (*s*), 68.67 (*d*), 63.69 (*t*), 62.96 (*t*), 34.30 (*t*), 32.74 (*t*), 29.17 (*t*), 29.16 (*t*), 29.03 (*t*), 27.73 (*t*), 25.65 (*t*), 24.97 (*t*), 24.92 (*t*), 18.07 (*t*). IR (cm^{-1}): 3620, 3307, 1726. MS (*m/z*): 255 (0.3). Analysis for $\text{C}_{15}\text{H}_{26}\text{O}_3$: C = 70.58%; H = 10.33% (calcd. C = 70.81%; H = 10.31%).

6-Hydroxyhexanoic acid 9-decynyl ester 15c: The product was obtained as a colourless oil in 0.99 g (97% yield). ^1H NMR (400 MHz, CDCl_3): δ 4.06 (*t*, J = 6.6, 2H); 3.65 (*m*, 2H); 2.32 (*t*, J = 7.6, 2H); 2.18 (*td*, J = 2.7/7.0, 2H); 1.94 (*t*, J = 2.6, 1H); 1.20-1.70 (*m*, 19H). ^{13}C NMR (100 MHz, CDCl_3): δ 173.77 (*s*), 84.67 (*s*), 68.09 (*d*), 64.62 (*t*), 62.66 (*t*), 34.24 (*t*), 32.32 (*t*), 29.06 (*t*), 28.93 (*t*), 28.60 (*t*), 28.40 (*t*), 25.85 (*t*), 25.28 (*t*), 24.66 (*t*), 18.35 (*t*). IR (cm^{-1}): 3625, 3307, 1724. MS (*m/z*): 269 (9). HR-MS: 268.2058 (calcd. for $\text{C}_{16}\text{H}_{28}\text{O}_3$: 268.2038). Analysis for $\text{C}_{16}\text{H}_{28}\text{O}_3$: C = 71.15%; H = 10.62% (calcd. C = 71.59%; H = 10.52%).

6-Hydroxyhexanoic acid 13-tetradecynyl ester 15d: The product was obtained as a white solid in 1.40 g (96% yield). ^1H NMR (400 MHz, CDCl_3): δ 4.06 (*t*, J = 6.8, 2H); 3.66 (*m*, 2H); 2.32 (*t*, J = 7.6, 2H); 2.18 (*td*, J = 2.7/7.1, 2H); 1.94 (*t*, J = 2.6, 1H); 1.48-1.70 (*m*, 8H); 1.20-1.45 (*m*, 19H). ^{13}C NMR (100 MHz, CDCl_3): δ 173.81 (*s*), 84.83 (*s*), 68.05 (*d*), 64.52 (*t*), 62.73 (*t*), 34.29 (*t*), 32.37 (*t*), 29.57 (*t*), 29.55 (*t*), 29.52 (*t*), 29.50 (*t*), 29.26 (*t*), 29.12 (*t*), 28.78 (*t*), 28.68 (*t*), 28.52 (*t*), 25.95 (*t*), 25.95 (*t*), 25.32 (*t*), 24.71 (*t*), 18.42 (*t*). IR (cm^{-1}): 3623, 3307, 1724. MS (*m/z*): 149 (4). HR-MS: 149.1372 (calcd. for $\text{C}_{11}\text{H}_{17}$: 149.1330). Analysis for $\text{C}_{20}\text{H}_{36}\text{O}_3$: C = 73.90%; H = 11.29% (calcd. C = 74.01%; H = 11.19%).

Parikh-Doering Oxidation of 15:

6-Oxohexanoic acid 5-hexynyl ester 16a: The product was obtained as a colourless oil in 0.52 g (83% yield). ^1H NMR (400 MHz, CDCl_3): δ 9.78 (*m*, 1H); 4.10 (*t*, $J = 6.4$, 2H); 2.44-2.51 (*m*, 2H); 2.30-2.40 (*m*, 2H); 2.24 (*td*, $J = 2.6/7.1$, 2H); 1.97 (*t*, $J = 2.6$, 1H); 1.50-1.80 (*m*, 8H). ^{13}C NMR (100 MHz, CDCl_3): δ 233.92 (*d*), 173.28 (*s*), 83.81 (*s*), 68.71 (*d*), 63.89 (*t*), 43.48 (*t*), 33.92 (*t*), 27.64 (*t*), 24.90 (*t*), 21.49 (*t*), 18.04 (*t*). IR (cm^{-1}): 3307, 1724. MS (m/z): 211 (2). HR-MS: 211.1337 (calcd. for $\text{C}_{12}\text{H}_{19}\text{O}_3$: 211.1334).

9-Oxononanoic acid 5-hexynyl ester 16b: The product was obtained as a colourless oil in 1.26 g (83% yield). ^1H NMR (200 MHz, CDCl_3): δ 9.76 (*t*, $J = 2.0$, 1H); 4.09 (*t*, $J = 6.6$, 2H); 2.42 (*dt*, $J = 2.0/7.4$, 2H); 2.30 (*t*, $J = 7.6$, 2H); 2.24 (*td*, $J = 2.8/7.0$, 2H); 1.96 (*t*, $J = 2.5$, 1H); 1.70-1.80 (*m*, 2H); 1.55-1.65 (*m*, 6H); 1.33 (*s*, 6H). ^{13}C NMR (100 MHz, CDCl_3): δ 202.70 (*d*), 173.76 (*s*), 68.68 (*d*), 63.71 (*t*), 43.84 (*t*), 34.25 (*t*), 28.90 (*t*), 27.72 (*t*), 24.96 (*t*), 24.86 (*t*), 21.98 (*t*), 18.08 (*t*). IR (cm^{-1}): 3307, 1713. MS (m/z): 252 (1). HR-MS: 252.1676 (calcd. for $\text{C}_{15}\text{H}_{24}\text{O}_3$: 252.1626).

6-Oxohexanoic acid 9-decynyl ester 16c: The product was obtained as a colourless oil in 0.79 g (90% yield). ^1H NMR (400 MHz, CDCl_3): δ 9.78 (*t*, $J = 1.5$, 1H); 4.06 (*t*, $J = 6.6$, 2H); 2.44-2.49 (*m*, 2H); 2.30-2.36 (*m*, 2H); 2.19 (*td*, $J = 2.6/7.0$, 2H); 1.94 (*t*, $J = 2.7$, 1H); 1.48-1.75 (*m*, 8H); 1.20-1.47 (*m*, 8H). ^{13}C NMR (100 MHz, CDCl_3): δ 201.99 (*d*), 173.32 (*s*), 84.68 (*s*), 68.11 (*d*), 64.54 (*t*), 43.50 (*t*), 33.99 (*t*), 29.07 (*t*), 28.95 (*t*), 28.61 (*t*), 28.42 (*t*), 25.86 (*t*), 24.41 (*t*), 21.53 (*t*), 18.37 (*t*). IR (cm^{-1}): 3307, 1724. MS (m/z): 269 (2). HR-MS: 268.2009 (calcd. for $\text{C}_{15}^{13}\text{CH}_{27}\text{O}_3$:

268.1993).

6-Oxohexanoic acid 13-tetradecynyl ester 16d: The product was obtained as a white solid in 0.96 g (75% yield), m.p. 31-33°C. ^1H NMR (400 MHz, CDCl_3): δ 9.77 (*t*, $J = 1.6$, 1H); 4.06 (*t*, $J = 6.9$, 2H); 2.43-2.50 (*m*, 2H); 2.30-2.37 (*m*, 2H); 2.18 (*td*, $J = 2.6/7.1$, 2H); 1.94 (*t*, $J = 2.7$, 1H); 1.48-1.70 (*m*, 8H); 1.25-1.43 (*m*, 16H). ^{13}C NMR (100 MHz, CDCl_3): δ 201.83 (*d*), 173.23 (*s*), 84.72 (*s*), 67.98 (*d*), 64.54 (*t*), 43.45 (*t*), 33.95 (*t*), 29.49 (*t*), 29.47 (*t*), 29.43 (*t*), 29.42 (*t*), 29.18 (*t*), 29.04 (*t*), 28.70 (*t*), 28.60 (*t*), 28.46 (*t*), 25.87 (*t*), 24.38 (*t*), 21.51 (*t*), 18.35 (*t*). IR (cm^{-1}): 3307, 1724. MS (*m/z*): 324 (0.2). HR-MS: 324.2656 (calcd. for $\text{C}_{19}^{13}\text{H}_{35}\text{O}_3$: 324.2630).

4-Pentynoic acid 10-hydroxydecyl ester 18a: The product was obtained as a yellow oil in 0.73 g (56% yield). ^1H NMR (400 MHz, CDCl_3): δ 4.10 (*t*, $J = 6.6$, 2H); 3.64 (*t*, $J = 6.6$, 2H); 2.47-2.58 (*m*, 4H); 1.98 (*t*, $J = 2.7$, 1H); 1.50-1.68 (*m*, 4H); 1.20-1.48 (*m*, 13H). ^{13}C NMR (100 MHz, CDCl_3): δ 171.86 (*s*), 82.58 (*s*), 68.96 (*d*), 63.08 (*t*), 33.45 (*t*), 32.81 (*t*), 29.49 (*t*), 29.42 (*t*), 29.38 (*t*), 29.19 (*t*), 28.60 (*t*), 25.88 (*t*), 25.73 (*t*), 14.43 (*t*). IR (cm^{-1}): 3622, 3308, 1728. MS (*m/z*): 256 (1). HR-MS: 255.1977 (calcd. for $\text{C}_{15}\text{H}_{27}\text{O}_3$: 255.1960). Analysis for $\text{C}_{15}\text{H}_{26}\text{O}_3$: C = 70.69%; H = 10.35% (calcd. C = 70.81%; H = 10.31%).

4-Pentynoic acid 11-hydroxyundecyl ester 18b: The product was obtained as a yellow oil in 1.17 g (51% yield). ^1H NMR (400 MHz, CDCl_3): δ 4.10 (*t*, $J = 6.6, 8$, 2H); 3.64 (*t*, $J = 6.6$, 2H); 2.46-2.58 (*m*, 4H); 1.97 (*t*, $J = 2.4$, 1H); 1.50-1.65 (*m*, 4H); 1.20-1.45 (*m*, 15H). ^{13}C NMR (100 MHz, CDCl_3): δ 171.85 (*s*), 82.54 (*s*), 68.93

(*d*), 64.89 (*t*), 63.06 (*t*), 33.40 (*t*), 32.77 (*t*), 29.53 (*t*), 29.44 (*t*), 29.38 (*t*), 29.18 (*t*), 28.57 (*t*), 25.85 (*t*), 25.71 (*t*), 14.39 (*t*). IR (cm^{-1}): 3620, 1730. MS (*m/z*): 269 (0.1). Analysis for $\text{C}_{16}\text{H}_{28}\text{O}_3$: C = 71.33%; H = 10.61% (calcd. C = 71.59%; H = 10.52%).

Parikh-Doering Oxidation of 18:

4-Pentynoic acid 10-oxodecyl ester 19a: The product was obtained as a yellow oil in 3.84 g (76% yield). ^1H NMR (400 MHz, CDCl_3): δ 9.75 (*t*, $J = 1.7$, 1H); 4.08 (*t*, $J = 6.6$, 2H); 2.46-2.56 (*m*, 4H); 2.41 (*td*, $J = 1.6/7.4$, 2H); 1.96 (*t*, $J = 2.5$, 1H); 1.56-1.65 (*m*, 4H); 1.28 (broad *s*, 10H). ^{13}C NMR (100 MHz, CDCl_3): δ 202.62 (*d*), 171.72 (*s*), 82.48 (*s*), 68.89 (*d*), 64.76 (*t*), 43.81 (*t*), 33.38 (*t*), 29.19 (*t*), 29.18 (*t*), 29.05 (*t*), 28.53 (*t*), 25.78 (*t*), 22.00 (*t*), 14.36 (*t*). IR (cm^{-1}): 3308, 1724. MS (*m/z*): 254 (13). HR-MS: 253.1857 (calcd. for $\text{C}_{15}\text{H}_{25}\text{O}_3$: 253.1803).

4-Pentynoic acid 11-Oxoundecyl ester 19b: The product was obtained as a colourless oil in 0.74 g (71% yield). ^1H NMR (400 MHz, CDCl_3): δ 9.77 (*t*, $J = 1.8$, 1H); 4.10 (*t*, $J = 6.8$, 2H); 2.48-2.63 (*m*, 4H); 2.42 (*td*, $J = 2.0/7.4$, 2H); 1.97 (*t*, $J = 2.4$, 1H); 1.50-1.75 (*m*, 4H); 1.20-1.40 (*m*, 12H). ^{13}C NMR (100 MHz, CDCl_3): δ 202.87 (*d*), 171.83 (*s*), 82.55 (*s*), 68.94 (*d*), 64.87 (*t*), 43.90 (*t*), 33.40 (*t*), 29.39 (*t*), 29.30 (*t*), 29.17 (*t*), 29.13 (*t*), 28.57 (*t*), 25.85 (*t*), 22.06 (*t*), 14.40 (*t*). IR (cm^{-1}): 3310, 1720. MS (*m/z*): 267 (0.4). HR-MS: 267.1947 (calcd. for $\text{C}_{16}\text{H}_{27}\text{O}_3$: 267.1960).

4-Pentynoic acid 10-bromodecyl ester 20a: The same procedure was used as for the synthesis of 7b. The product was obtained as a clear colourless oil in 1.55 g (81% yield). ^1H NMR (400 MHz, CDCl_3): δ 4.10 (*t*, $J = 6.6$, 2H); 3.41 (*t*, $J = 6.9$,

2H); 2.48-2.58 (*m*, 4H); 1.98 (*t*, $J = 2.5$, 1H); 1.86 (*m*, 2H); 1.60 (*m*, 2H); 1.28-1.45 (*m*, 12H). ^{13}C NMR (100MHz, CDCl_3): δ 171.77 (*s*), 82.53 (*s*), 68.91 (*d*), 64.84 (*t*), 33.88 (*t*), 33.43 (*t*), 32.81 (*t*), 29.34 (*t*), 29.30 (*t*), 29.14 (*t*), 28.70 (*t*), 28.58 (*t*), 28.13 (*t*), 25.84 (*t*), 14.40 (*t*). IR (cm^{-1}): 1736. MS (m/z): 265 (0.3).

Oxidation of 20a: To a mixture of anhydrous trimethylamine N-oxide (0.3 g, 4 mmol) in DMSO (1 mL) was added a solution of **12a** (0.32 g, 1 mmol) in DMSO (1 mL) at room temperature. After stirring 5 h at room temperature, the reaction mixture was poured into ice-cold brine and extracted with ether. The combined ether extracts were washed with brine and dried (MgSO_4). After concentration and flash chromatography (SiO_2 ; *n*-hexane : ether = 3/1) the aldehyde was afforded as a clear colourless oil in 0.15 g (60% yield).

(1*S*,2*E*)-2-Cyclotridecenol 25a: The product was obtained as a clear colourless oil in 0.04 g (20% yield). GC: $t_r = 1.15$ min. (for **25a**) and $t_r = 0.90$ min. (for **26a**) (160°C). ^1H NMR (400 MHz, CDCl_3): δ 5.58 (*ddd*, $J = 5.3/9.3/15.0$, 1H); 5.40 (*dd*, $J = 8.0/14.6$, 1H); 4.07 (*m*, 1H); 2.15 (*m*, 1H); 2.00 (*m*, 1H); 1.75 (*m*, 1H); 1.00-1.60 (*m*, 18H). ^{13}C NMR (100 MHz, CDCl_3): δ 134.07 (*d*), 133.36 (*d*), 73.60 (*d*), 26.04 (*t*), 31.06 (*t*), 27.26 (*t*), 26.36 (*t*), 25.91 (*t*), 25.84 (*t*), 25.84 (*t*), 25.01 (*t*), 24.93 (*t*), 22.45 (*t*). IR (cm^{-1}): 3604. MS (m/z): 196 (4). HR-MS: 196.1810 (calcd. for $\text{C}_{13}\text{H}_{24}\text{O}$: 196.1827). Enantiomeric purity of 90% ee by ^{19}F NMR of the (1*R*)-MTPA esters: 92.14 (*S*), 92.05 (*R*).

(1*S*,2*E*)-2-Cyclotetradecenol 25b: The product was obtained as a semi-solid in 0.07 g (35% yield). GC: *rt* = 1.61 min. (for **25b**) and *rt* = 1.30 min. (for **26b**) (160°C). $[\alpha]_D = -29.2$ (*c* = 0.633, CHCl₃). ¹H NMR (400 MHz, CDCl₃): δ 5.53 (*ddd*, *J* = 4.9/9.3/15.0, 1H); 5.37 (*dd*, *J* = 8.0/15.0, 1H); 4.08 (*m*, 1H); 2.20 (*m*, 1H); 2.00 (*m*, 1H); 1.65 (*m*, 1H); 0.90-1.55 (*m*, 20H). ¹³C NMR (100 MHz, CDCl₃): δ 134.23 (*d*), 133.14 (*d*), 73.69 (*d*), 35.90 (*t*), 31.31 (*t*), 27.23 (*t*), 26.41 (*t*), 26.25 (*t*), 25.19 (*t*), 25.07 (*t*), 24.25 (*t*), 23.48 (*t*), 22.03 (*t*). IR (cm⁻¹): 3604. MS (*m/z*): 210 (11). HR-MS: 210.1981 (calcd. for C₁₄H₂₆O: 210.1984). Enantiomeric purity of 91% ee by ¹⁹F NMR of the (1*R*)-MTPA esters: 92.12 (*S*), 92.04 (*R*).

(1*S*,2*E*)-2-Cyclopentadecenol 25c: The product was obtained as a semi-solid in 0.13 g (60% yield). GC: *rt* = 3.30 min. (for **25c**) and *rt* = 2.69 min. (for **26c**) (160°C). $[\alpha]_D = -33.4$ (*c* = 1.01, CHCl₃). ¹H NMR (400 MHz, CDCl₃): δ 5.59 (*ddd*, *J* = 4.9/8.8/15.5, 1H); 5.43 (*dd*, *J* = 7.5/15.5, 1H); 4.08 (*m*, 1H); 2.20 (*m*, 1H); 1.65-1.80 (*m*, 1H); 1.10-1.60 (*m*, 22H). ¹³C NMR (100 MHz, CDCl₃): δ 133.72 (*d*), 133.69 (*d*), 73.57 (*d*), 36.93 (*t*), 31.32 (*t*), 28.17 (*t*), 27.50 (*t*), 27.14 (*t*), 27.02 (*t*), 26.90 (*t*), 26.85 (*t*), 26.81 (*t*), 25.78 (*t*), 24.10 (*t*). IR (cm⁻¹): 3605. MS (*m/z*): 224 (15). HR-MS: 224.2132 (calcd. for C₁₅H₂₈O: 224.2140). Enantiomeric purity of 88% ee by ¹⁹F NMR of the (1*R*)-MTPA esters: 92.15 (*S*), 92.06 (*R*).

(1*S*,2*E*)-2-Cyclooctadecenol 25d: The product was obtained as a clear colourless oil in 0.17 g (61% yield). GC: *rt* = 6.68 min. (for **25d**) and *rt* = 6.26 min. (for **26d**) (160°C, 5min., 10°C/min., 270°C). $[\alpha]_D = -32.0$ (*c* = 0.959, CHCl₃). ¹H NMR (400 MHz, CDCl₃): δ 5.60 (*m*, 1H); 5.41 (*dd*, *J* = 7.7/15.3, 1H); 4.06 (*m*, 1H);

1.90-2.20 (*m*, 2H); 1.60 (*m*, 1H); 1.20-1.55 (*m*, 28H). ^{13}C NMR (100 MHz, CDCl_3): δ 133.39 (*d*), 132.62 (*d*), 73.55 (*d*), 37.02 (*t*), 31.86 (*t*), 28.60 (*t*), 28.41 (*t*), 28.28 (*t*), 27.80 (*t*), 27.63 (*t*), 27.49 (*t*), 27.43 (*t*), 26.90 (*t*), 26.53 (*t*), 24.50 (*t*). IR (cm^{-1}): 3606. MS (*m/z*): 266 (35). HR-MS: 266.2623 (calcd. for $\text{C}_{18}\text{H}_{34}\text{O}$: 266.2610). Enantiomeric purity of 91% ee by ^{19}F NMR of the (1*R*)-MTPA esters: 92.18 (*S*), 92.12 (*R*).

(1*S*,2*E*)-2-Cycloheñicosenol 25e: The product was obtained as a semi-solid in 0.07 g (43% yield). GC: *rt* = 10.24 min. (for **25e**) and *rt* = 10.04 min. (for **26e**) (160°C, 5 min., 10°C/min., 270°C). $[\alpha]_{\text{D}} = -17.0$ (*c* = 0.959, CHCl_3). ^1H NMR (400 MHz, CDCl_3): δ 5.61 (*ddd*, *J* = 5.8/8.0/15.0, 1H); 5.42 (*dd*, *J* = 7.5/15.5, 1H); 4.05 (*m*, 1H); 1.90-2.20 (*m*, 2H); 1.20-1.60 (*m*, 35H). ^{13}C NMR (100 MHz, CDCl_3): δ 133.20 (*d*), 132.69 (*d*), 73.51 (*d*), 37.15 (*t*), 31.96 (*t*), 29.33 (*t*), 29.20 (*t*), 28.94 (*t*), 28.67 (*t*), 28.55 (*t*), 28.36 (*t*), 28.33 (*t*), 28.30 (*t*), 28.08 (*t*), 27.93 (*t*), 27.67 (*t*), 24.90 (*t*). IR (cm^{-1}): 3606. MS (*m/z*): 308 (2). HR-MS: 308.3061 (calcd. for $\text{C}_{21}\text{H}_{40}\text{O}$: 308.3079). Enantiomeric purity of 88% ee by ^{19}F NMR of the (1*R*)-MTPA esters: 92.19 (*S*), 92.12 (*R*).

(7*S*,8*E*)-7-Hydroxyoxacyclotridec-8-en-2-one 27a: The product was obtained as a clear colourless oil in 0.03 g (14% yield). $[\alpha]_{\text{D}} = -32.6$ (*c* = 0.5, CH_2Cl_2). ^1H NMR (400 MHz, CDCl_3): δ 5.65-5.78 (*m*, 1H); 5.35 (*dd*, *J* = 7.4/15.1, 1H); 4.18-4.25 (*m*, 1H); 4.03-4.13 (*m*, 2H); 2.23-2.38 (*m*, 2H); 1.94-2.12 (*m*, 2H); 1.35-1.80 (*m*, 10H); 1.20-1.30 (*m*, 1H). ^{13}C NMR (100 MHz, CDCl_3): δ 173.54 (*s*), 133.37 (*d*), 133.19 (*d*), 73.27 (*d*), 64.25 (*t*), 35.11 (*t*), 34.80 (*t*), 30.48 (*t*), 27.11 (*t*), 27.00 (*t*), 24.63 (*t*), 24.63 (*t*), 23.58 (*t*). IR (cm^{-1}): 3606, 1722. MS (*m/z*): 212 (0.2). HR-MS:

212.1412 (calcd. for $C_{12}H_{20}O_3$: 212.1413). Enantiomeric purity of 71% ee by ^{19}F NMR of the (1*R*)-MTPA esters: 92.13 (*S*), 92.07 (*R*).

(10*S*,11*E*)-10-Hydroxyoxacyclohexadec-11-en-2-one 27b: The product was obtained as a clear colourless oil in 0.17 g (65% yield). $[\alpha]_D = -43.1$ ($c = 1.22$, CH_3Cl). 1H NMR (400 MHz, $CDCl_3$): δ 5.52-5.61 (*m*, 1H); 5.38 (*dd*, $J = 7.4/15.4$, 1H); 4.14-4.23 (*m*, 1H); 4.02-4.10 (*m*, 2H); 2.29-2.40 (*m*, 2H); 2.15-2.28 (*m*, 1H); 2.00-2.12 (*m*, 1H); 1.82-1.91 (*m*, 1H); 1.10-1.80 (*m*, 16H). ^{13}C NMR (100 MHz, $CDCl_3$): δ 173.84 (*s*), 134.27 (*d*), 131.90 (*d*), 73.54 (*d*), 63.76 (*t*), 36.69 (*t*), 34.67 (*t*), 31.64 (*t*), 28.26 (*t*), 28.24 (*t*), 27.99 (*t*), 27.11 (*t*), 25.35 (*t*), 24.89 (*t*), 23.81 (*t*). IR (cm^{-1}): 3604, 1723. MS (m/z): 254 (1). HR-MS: 254.1790 (calcd. for $C_{14}^{13}CH_{25}O_3$: 254.1837). Enantiomeric purity of 89% ee by ^{19}F NMR of the (1*R*)-MTPA esters: 92.13 (*S*), 92.08 (*R*).

(7*S*,8*E*)-7-Hydroxyoxacycloheptadec-8-en-2-one 27c: The product was obtained as a clear colourless oil in 45% yield. $[\alpha]_D = -37.4$ ($c = 0.5$, CH_3Cl). 1H NMR (400 MHz, $CDCl_3$): δ 5.50-5.60 (*m*, 1H); 5.40 (*dd*, $J = 7.4/15.3$, 1H); 4.11-4.21 (*m*, 2H); 4.02-4.10 (*m*, 1H); 2.27-2.43 (*m*, 2H); 2.20-2.25 (*m*, 1H); 1.85-2.05 (*m*, 1H); 1.10-1.74 (*m*, 19H). ^{13}C NMR (100 MHz, $CDCl_3$): δ 172.94 (*s*), 134.46 (*d*), 131.71 (*d*), 73.20 (*d*), 63.28 (*t*), 37.64 (*t*), 34.39 (*t*), 31.92 (*t*), 29.22 (*t*), 28.88 (*t*), 28.63 (*t*), 28.36 (*t*), 27.10 (*t*), 25.43 (*t*), 25.16 (*t*), 24.65 (*t*). IR (cm^{-1}): 3602, 1724. MS (m/z): 268 (1). HR-MS: 268.2037 (calcd. for $C_{16}H_{28}O_3$: 268.2039). Enantiomeric purity of 90% ee by ^{19}F NMR of the (1*R*)-MTPA esters: 92.19 (*S*), 92.13 (*R*).

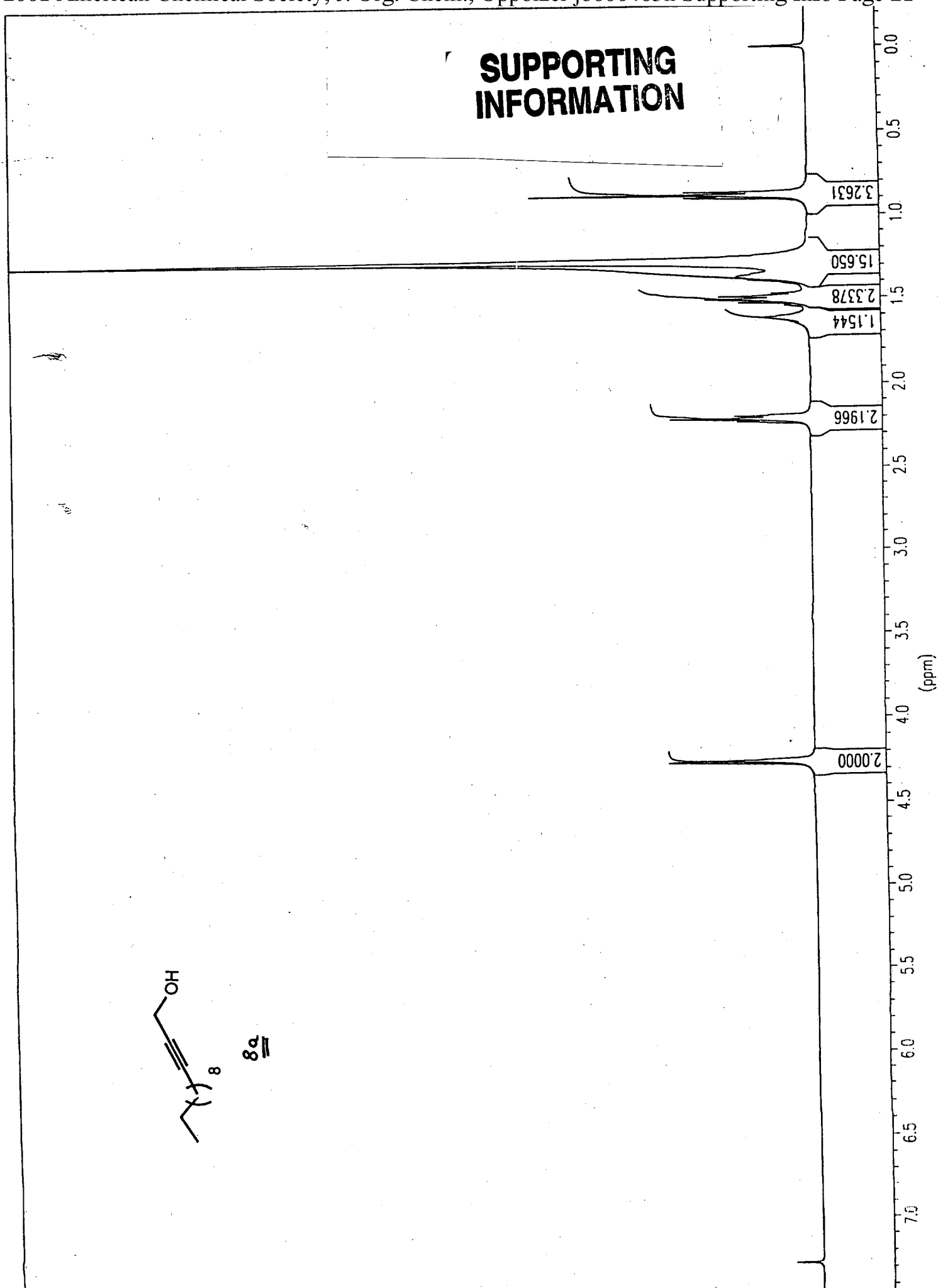
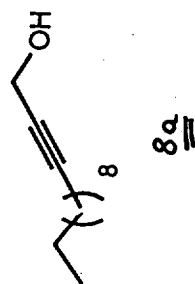
(7*S*,8*E*)-7-Hydroxyoxacycloheneicos-8-en-2-one 27d: The product was obtained as a clear colourless oil in 48% yield. $[\alpha]_D = -18.3$ ($c = 1.03$, CH_3Cl). ^1H NMR (400 MHz, CDCl_3): δ 5.42 (*m*, 1H); 5.42 (*dd*, $J = 7.4/15.1$, 1H); 4.00-4.13 (*m*, 3H); 2.31 (*t*, $J = 7.4$, 2H); 1.95-2.14 (*m*, 2H); 1.58-1.61 (*m*, 6H); 1.20-1.55 (*m*, 21H). ^{13}C NMR (100 MHz, CDCl_3): δ 173.71 (*s*), 133.20 (*d*), 132.66 (*d*), 73.07 (*d*), 64.38 (*t*), 36.85 (*t*), 34.42 (*t*), 31.59 (*t*), 28.62 (*t*), 28.46 (*t*), 28.37 (*t*), 28.27 (*t*), 28.26 (*t*), 28.21 (*t*), 28.15 (*t*), 27.96 (*t*), 27.81 (*t*), 25.70 (*t*), 24.97 (*t*), 24.69 (*t*). IR (cm^{-1}): 3604, 1723. MS (m/z): 324 (19). HR-MS: 324.2696 (calcd. for $\text{C}_{20}\text{H}_{36}\text{O}_3$: 324.2665). Enantiomeric purity of 90% ee by ^{19}F NMR of the (1*R*)-MTPA esters: 92.26 (*S*), 92.19 (*R*).

(5*E*,7*S*)-7-Hydroxyoxacyclohexadec-5-en-2-one 29a: The product was obtained as a clear colourless oil in 65% yield. $[\alpha]_D = -21.4$ ($c = 0.52$, CH_3Cl). ^1H NMR (400 MHz, CDCl_3): δ 5.58-5.70 (*m*, 1H); 5.48 (*dd*, $J = 7.4/15.4$, 1H); 4.17-4.25 (*m*, 1H); 4.00-4.15 (*m*, 2H); 2.30-2.50 (*m*, 4H); 1.60-1.78 (*m*, 5H); 1.15-1.58 (*m*, 12H). ^{13}C NMR (100 MHz, CDCl_3): δ 172.22 (*s*), 135.11 (*d*), 129.79 (*d*), 72.60 (*d*), 63.33 (*t*), 36.59 (*t*), 34.18 (*t*), 28.05 (*t*), 27.57 (*t*), 27.13 (*t*), 26.91 (*t*), 26.84 (*t*), 26.21 (*t*), 24.36 (*t*), 24.34 (*t*). IR (cm^{-1}): 3604, 1727. MS (m/z): 237 (9). HR-MS: 237.1866 (calcd. for $\text{C}_{15}\text{H}_{25}\text{O}_2$: 237.1855). Enantiomeric purity of 90% ee by ^{19}F NMR of the (1*R*)-MTPA esters: 92.15 (*S*), 92.11 (*R*).

(5*E*,7*S*)-7-Hydroxyoxacycloheptadec-5-en-2-one 29b: The product was obtained as a clear colourless oil in 44% yield. $[\alpha]_D = -17.8$ ($c = 1.25$, CH_3Cl). ^1H NMR (400 MHz, CDCl_3): δ 5.62-5.71 (*m*, 1H); 5.51 (*dd*, $J = 7.2/15.4$, 1H); 4.15-4.23

(*m*, 1H); 4.00-4.11 (*m*, 2H); 2.30-2.45 (*m*, 4H); 1.40-1.70 (*m*, 6H); 1.15-1.38 (*m*, 12H). ^{13}C NMR (100 MHz, CDCl_3): δ 172.37 (*s*), 135.36 (*d*), 129.84 (*d*), 72.73 (*d*), 63.83 (*t*), 37.01 (*t*), 34.18 (*t*), 28.62 (*t*), 28.05 (*t*), 27.95 (*t*), 27.66 (*t*), 27.43 (*t*), 27.26 (*t*), 26.65 (*t*), 24.81 (*t*), 24.13 (*t*). IR (cm^{-1}): 3604, 1726. MS (*m/z*): 268 (3). HR-MS: 268.2045 (calcd. for $\text{C}_{16}\text{H}_{28}\text{O}_3$: 268.2038). Enantiomeric purity of 73% ee by ^{19}F NMR of the (1*R*)-MTPA esters: 92.19 (*S*), 92.14 (*R*).

SUPPORTING INFORMATION



13C CPD PULS

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PROCNO 1

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D1 1.0000000 sec
S2 16 dB
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DE 25.0 usec
SF01 100.6248040 MHz
SFO1 25000.00 Hz
TD 65536
NS 636
DS 2

F2 - Processing parameters
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GB 0
PC 1.00

1D NMR plot parameters
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F1P 225.000 ppm
F1 22638.12 Hz
F2P -10.000 ppm
F2 -1006.14 Hz
PPHCH 7.28662 ppm/ci
HZCH 733.15327 Hz/cm

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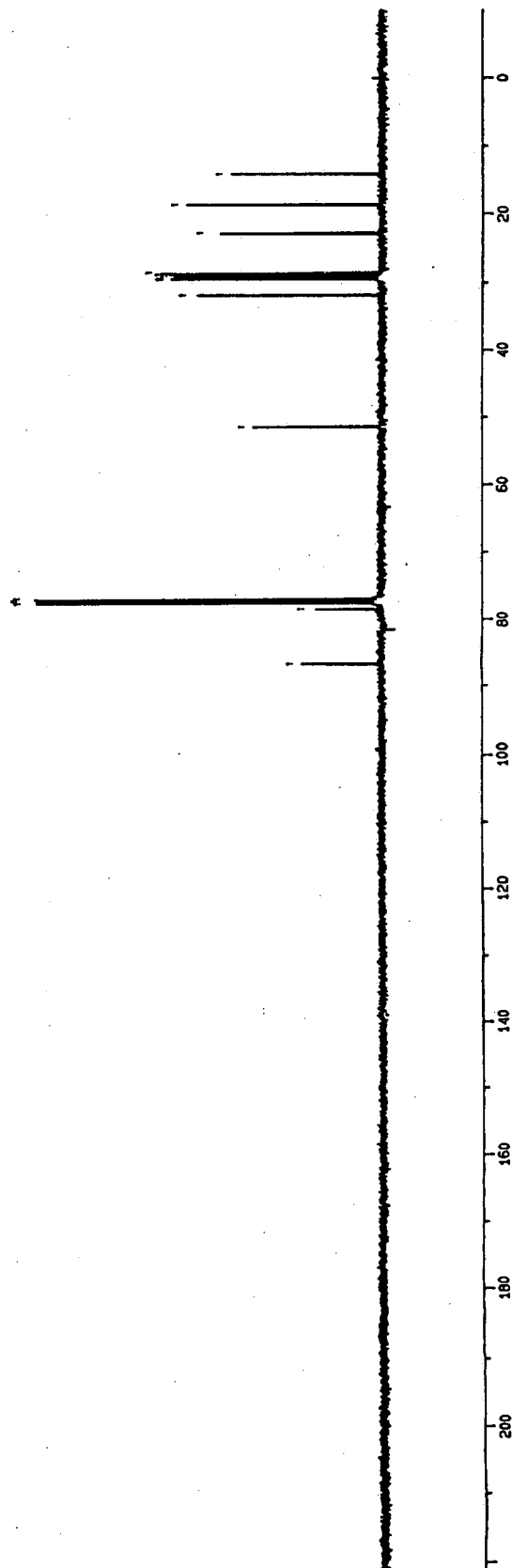
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8a

22





8c

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2.1906
2.1961
2.2038
2.2082
2.2138
2.2204
2.2259
2.2315

4.2410
4.2454
4.2509
4.2553
4.2609
4.2664

23

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(ppm)

2.20
(ppm)

4.26
(ppm)

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2.0371

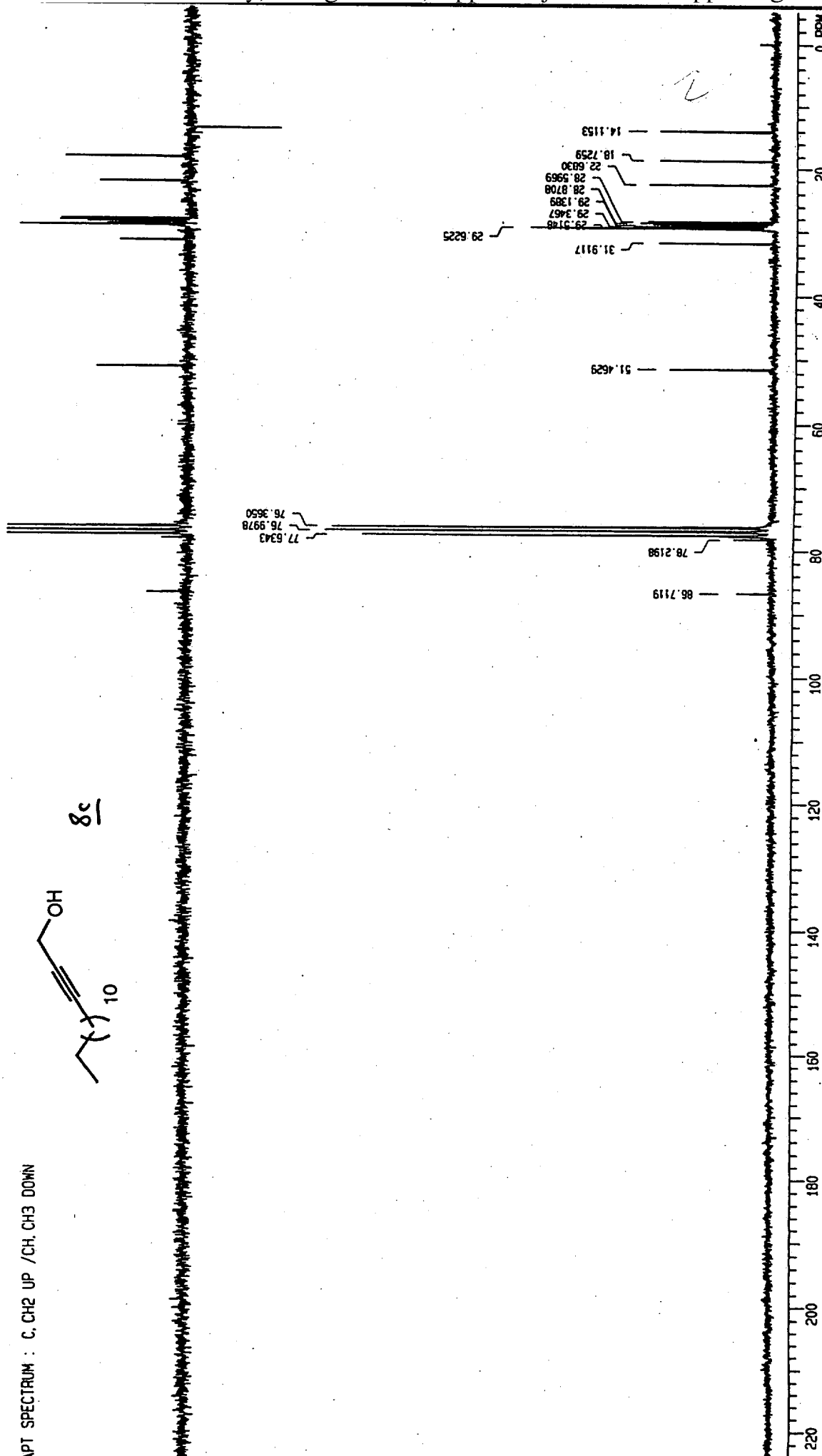
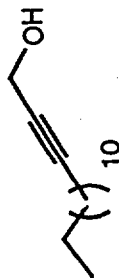
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1.9619

1.8375

(ppm)

APT SPECTRUM : C, CH2 UP /CH, CH3 DOWN



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 File C
 Date 08-20-96
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 C13-63M JPS

Pulse Sequence _____
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 Temp _____ °C
 Solvent CDCL3

EXPERIMENT

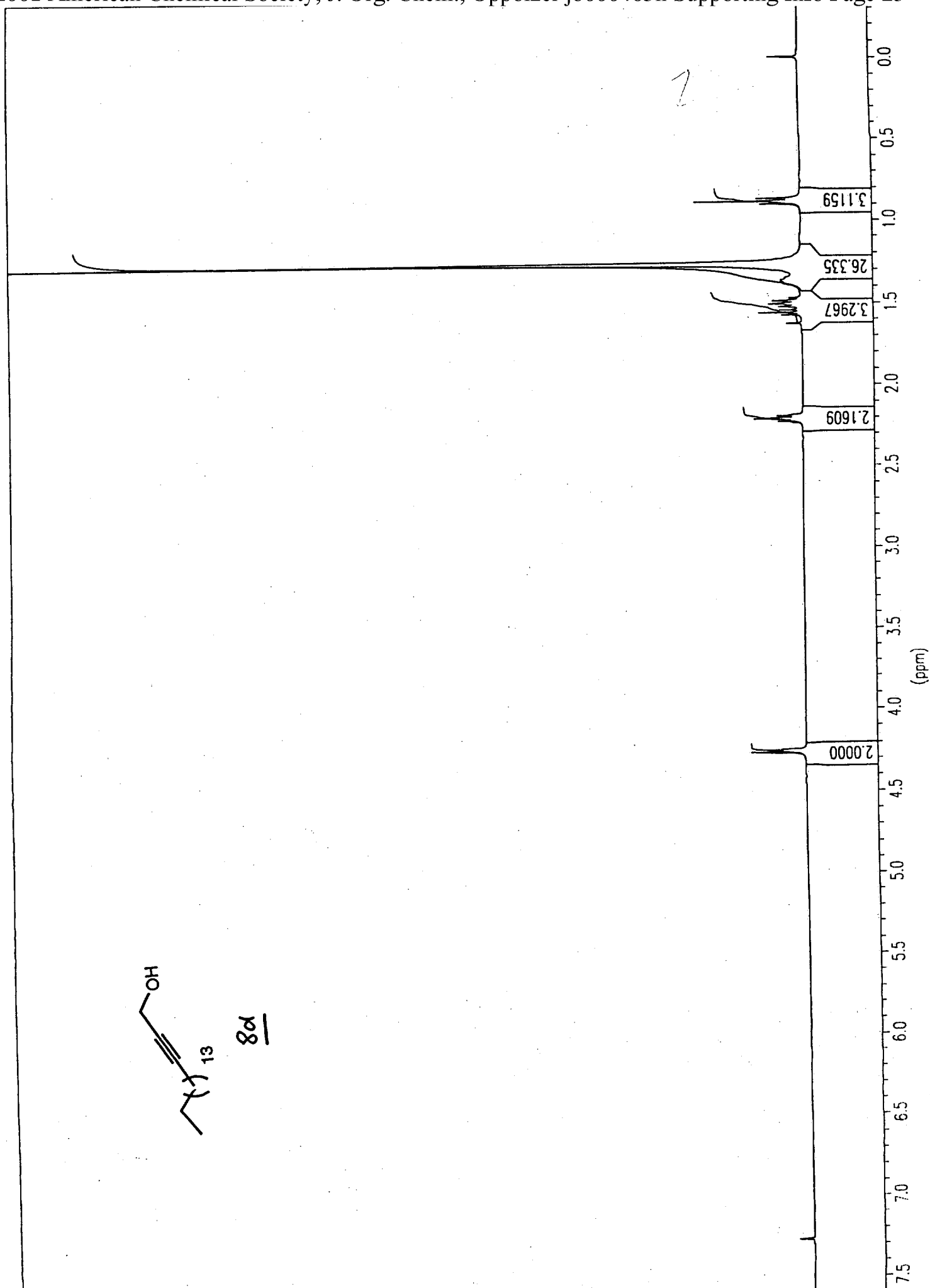
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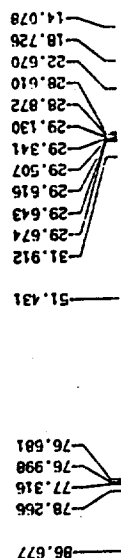
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DECOUPLE

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 Offset 7.1 sec





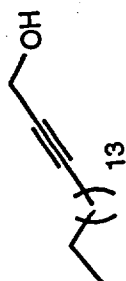
13C CPD PULS

Current Data Parameters
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EXPNO 20
PROCNO 1

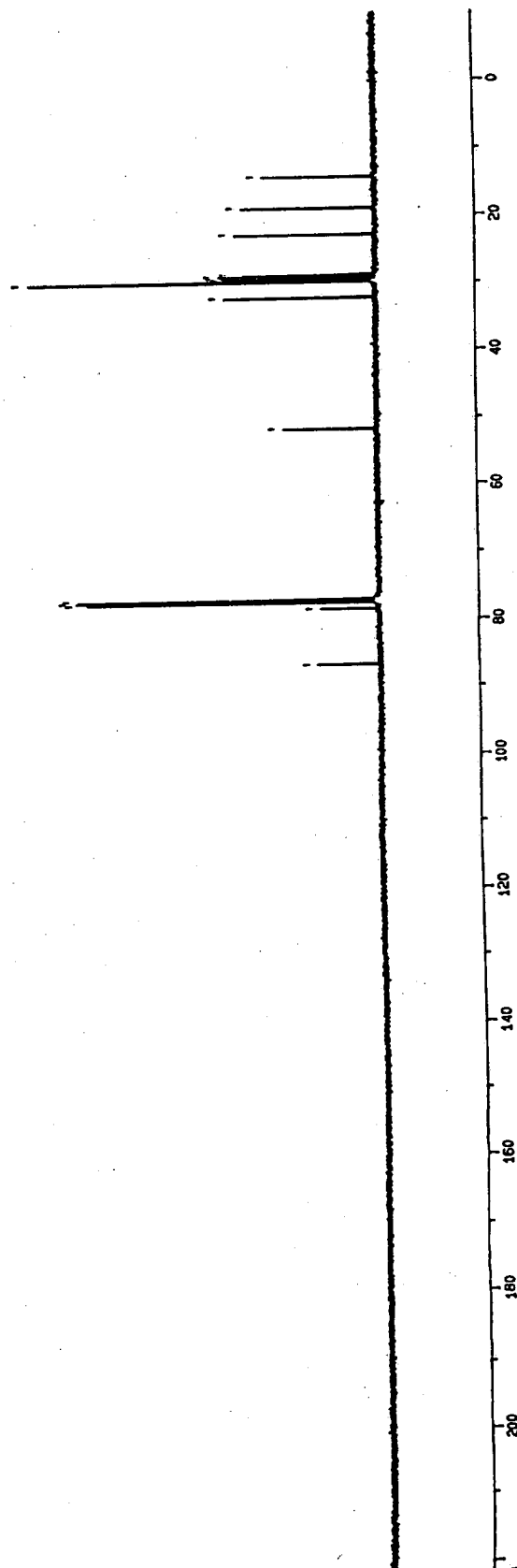
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RG 32768
NUCLEUS 13C
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D1 1.0000000 sec
S2 16 dB
P1 6.7 usec
DE 25.0 usec
SF01 100.6248040 MHz
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TD 65536
NS 983
DS 2

F2 - Processing parameters
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WDW EM
SSB 0
LB 1.00 Hz
GB 0
PC 1.00

1D NMR plot parameters
CX 32.25 cm
FIP 225.000 ppm
F1 22638.12 Hz
F2 -10.000 ppm
F2 -1006.14 Hz
PPMCH 7.28682 ppm/c
HZCH 733.15327 Hz/cm



8a



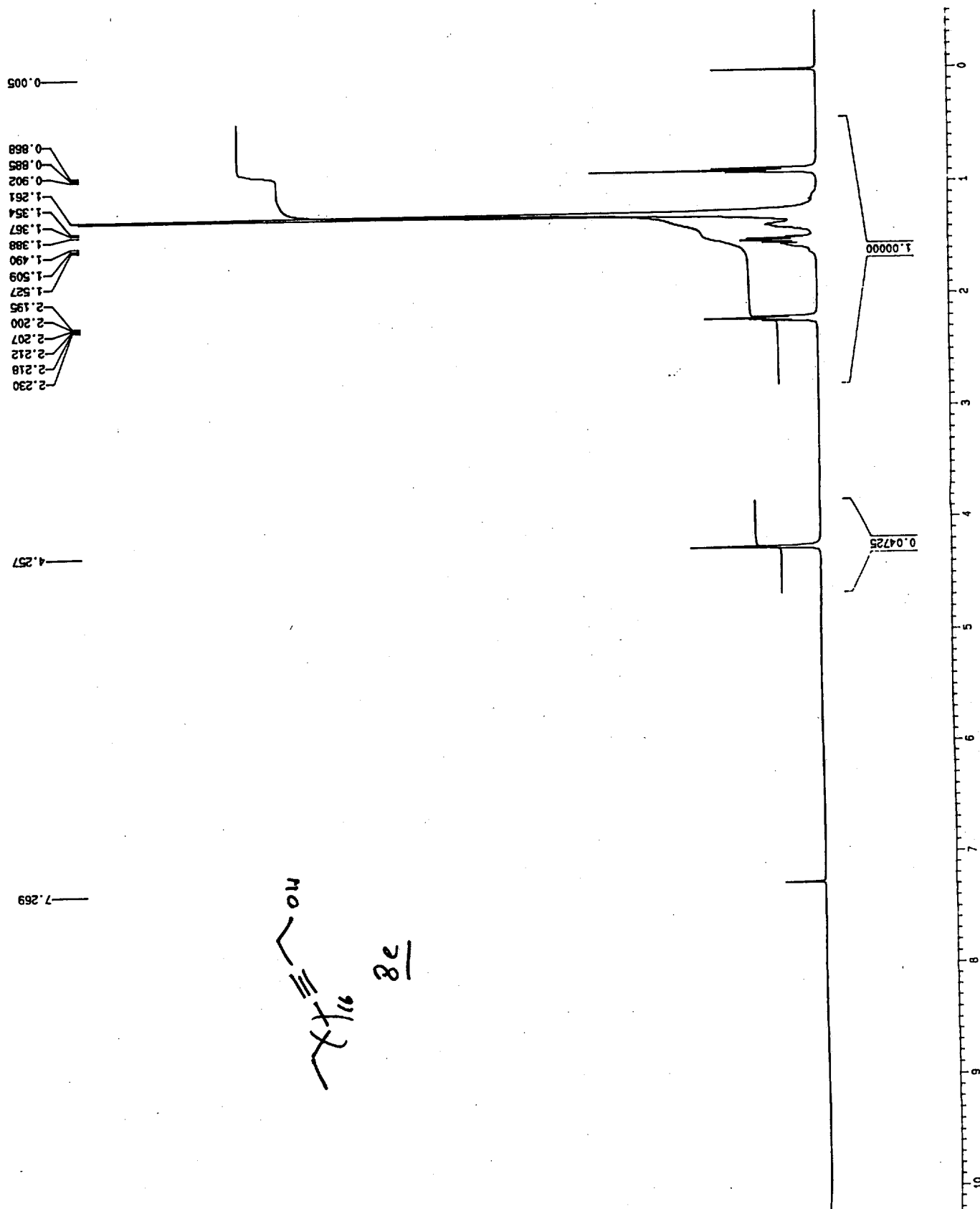
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1H SPECTRUM

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EXPNO 10
PROCNO 1

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FIDRES 0.246110 Hz
DQ 62.0 u
R6 128
NUCLEUS 1H
HL1 2 d
D1 3.0000000 s
P1 14.6 u
DE 88.6 u
SF01 400.1360000 MHz
SWH 8064.32 Hz
TD 32768
NS 64
DS 0

F2 - Processing parameter
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SF 400.1343904 MHz
WDW EM
SSB 0
LB 0.40 Hz
GB 0
PC 4.00

1D NMR plot parameters
CX 32.00 Hz
FJP 10.500 Hz
F1 4201.41 Hz
F2P -0.500 Hz
F2 -200.07 Hz



54463

13C CPD PULS

Current Data Parameters
NAME EMEB5
EXPNO 11
PROCNO 1

F2 - Acquisition Parameters

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Time 13.27
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SOLVENT CDCl3
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FIDRES 0.381470
DQ 20.0
R6 32768
NUCLEUS 13C
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D1 1.0000000
S2 16
P1 6.7
DE 25.0
SF01 100.6248040
SHH 25000.00
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F2 - Processing parameters

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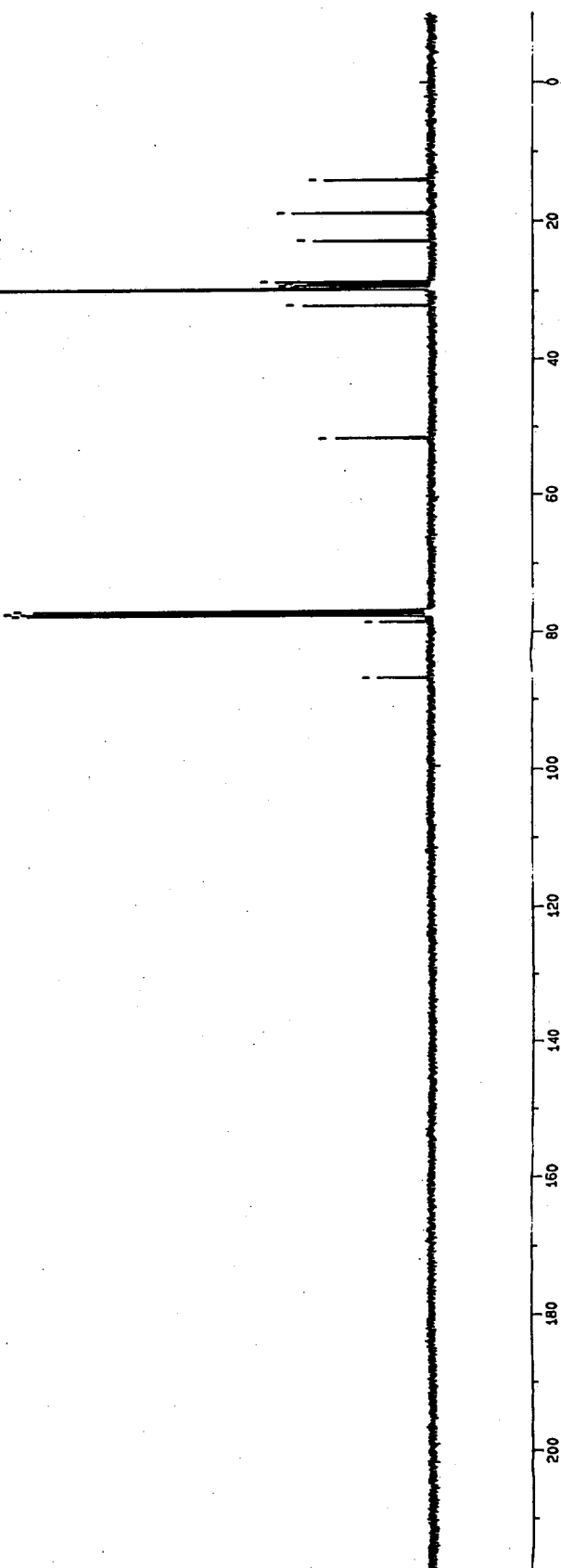
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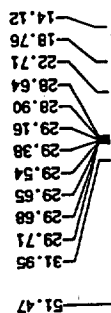
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76.993
77.311
78.289
-86.684



8c

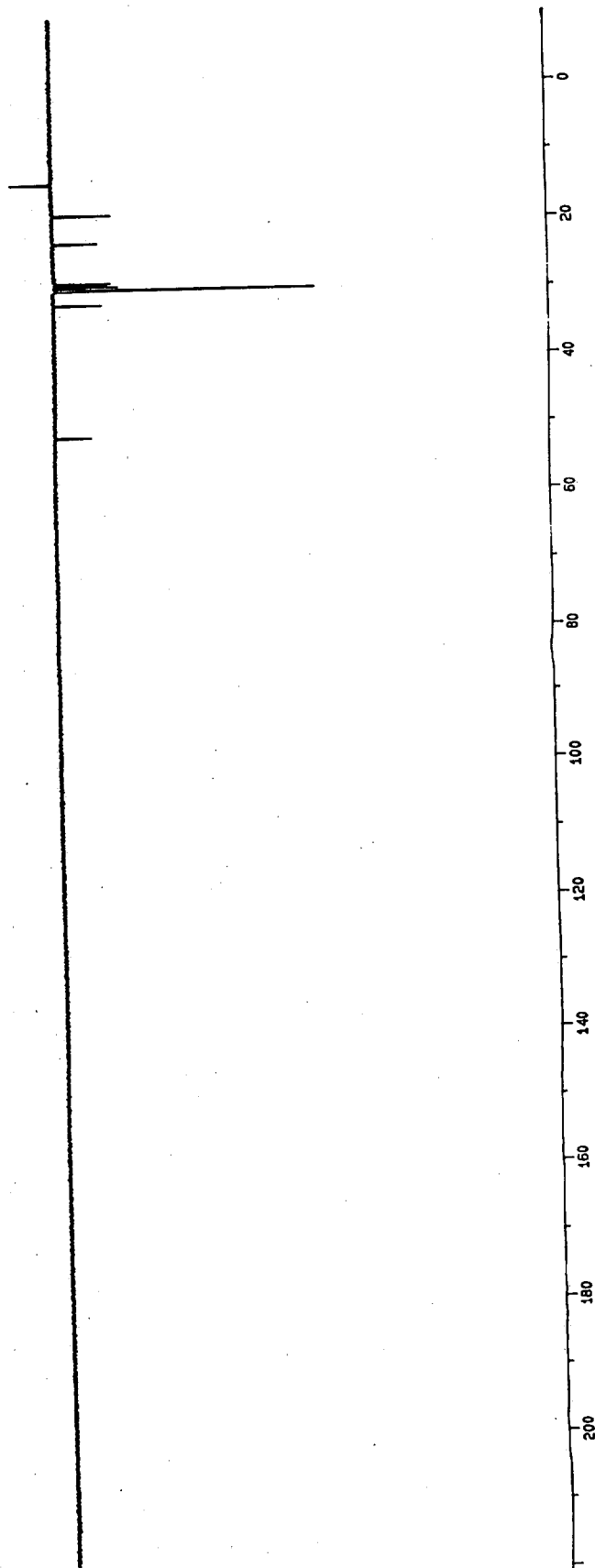


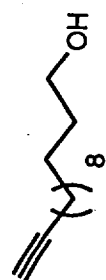
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CH2 DOWN



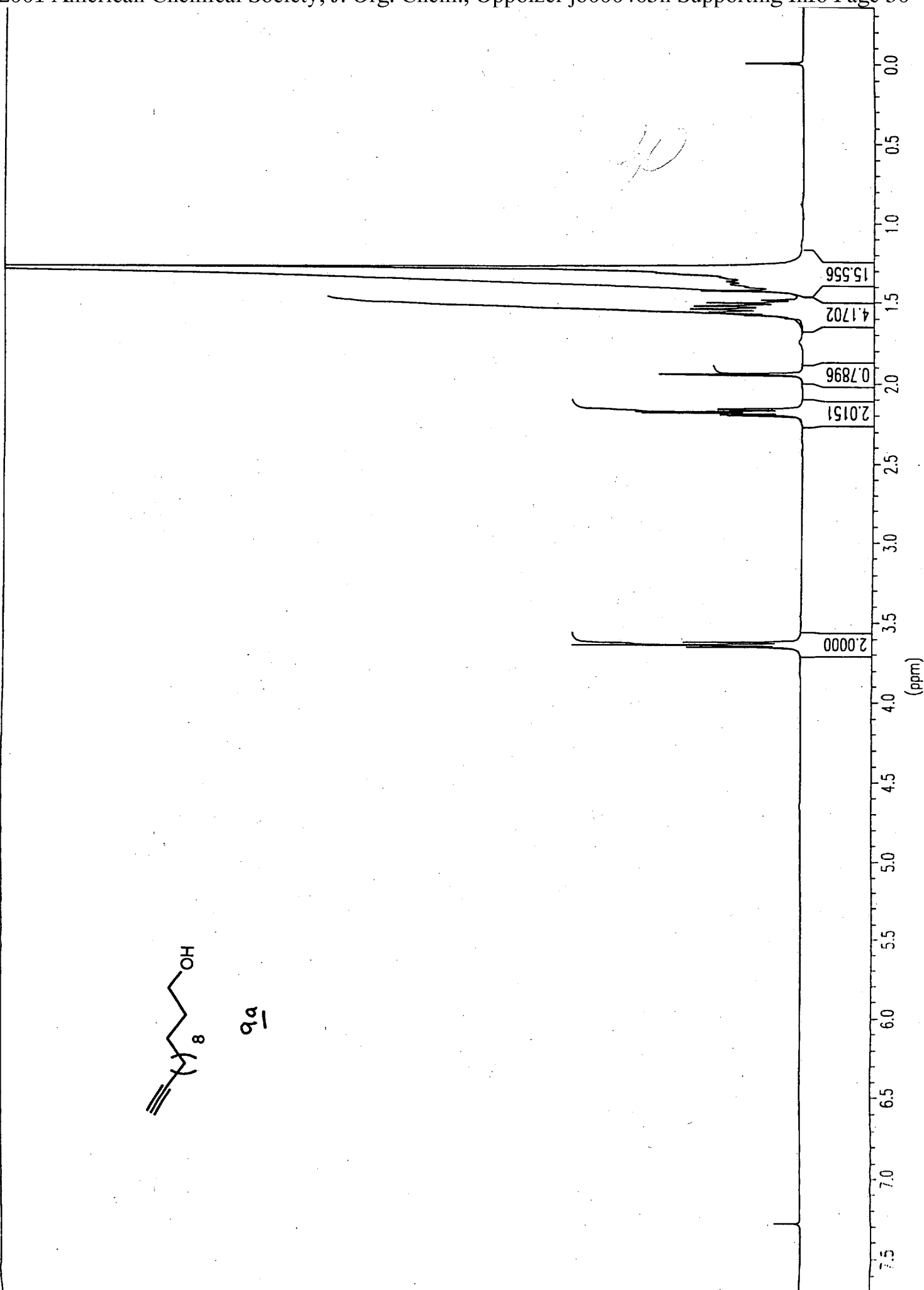
18

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R6	32768
NUCLEUS	
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S1	2
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P1	6.7 Hz
P2	13.4 Hz
S2	16 Hz
DE	25.0 Hz
SF01	100.6248040 MHz
SMH	25000.00 MHz
TD	65536
NS	500
DS	4
F2 - Processing parameters	
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SF	100.6138693 MHz
WDW	EM
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GB	0
PC	1.00
1D NMR plot parameters	
CX	32.25
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F1	22638.12
F2P	-10.000
F2	-1006.14





9a



13C CPD PULS

Current Data Parameters
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EXPNO 11
PROCNO 1

F2 - Acquisition Parameters
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NUCLEUS 13C
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S2 16 dB
P1 6.7 usec
DE 25.0 usec
SF01 100.6248040 MHz
SWH 25000.00 Hz
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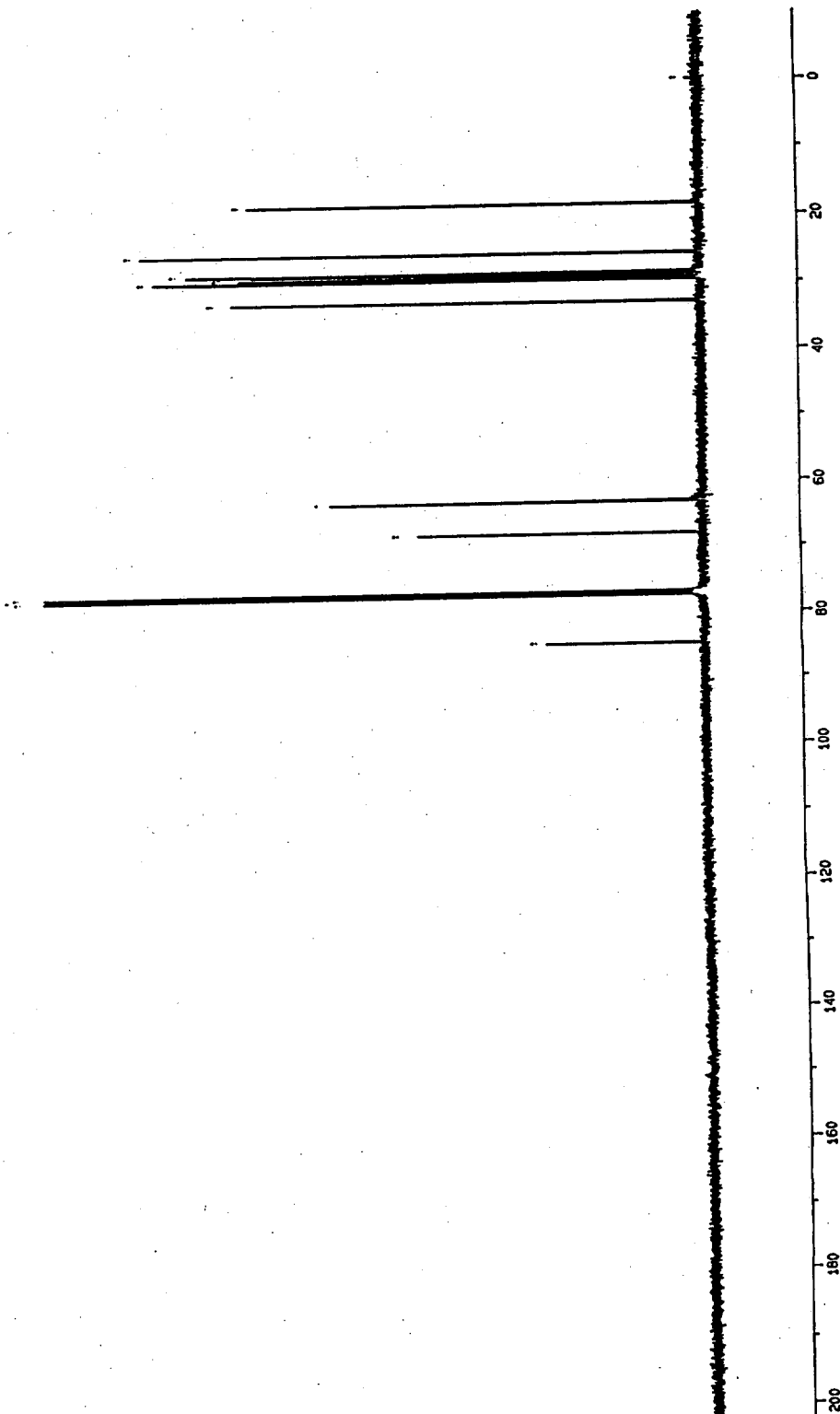
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F2 -10.000 ppa
F2 -1006.14 Hz
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HZCH 733.15533 Hz/cm

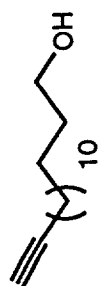
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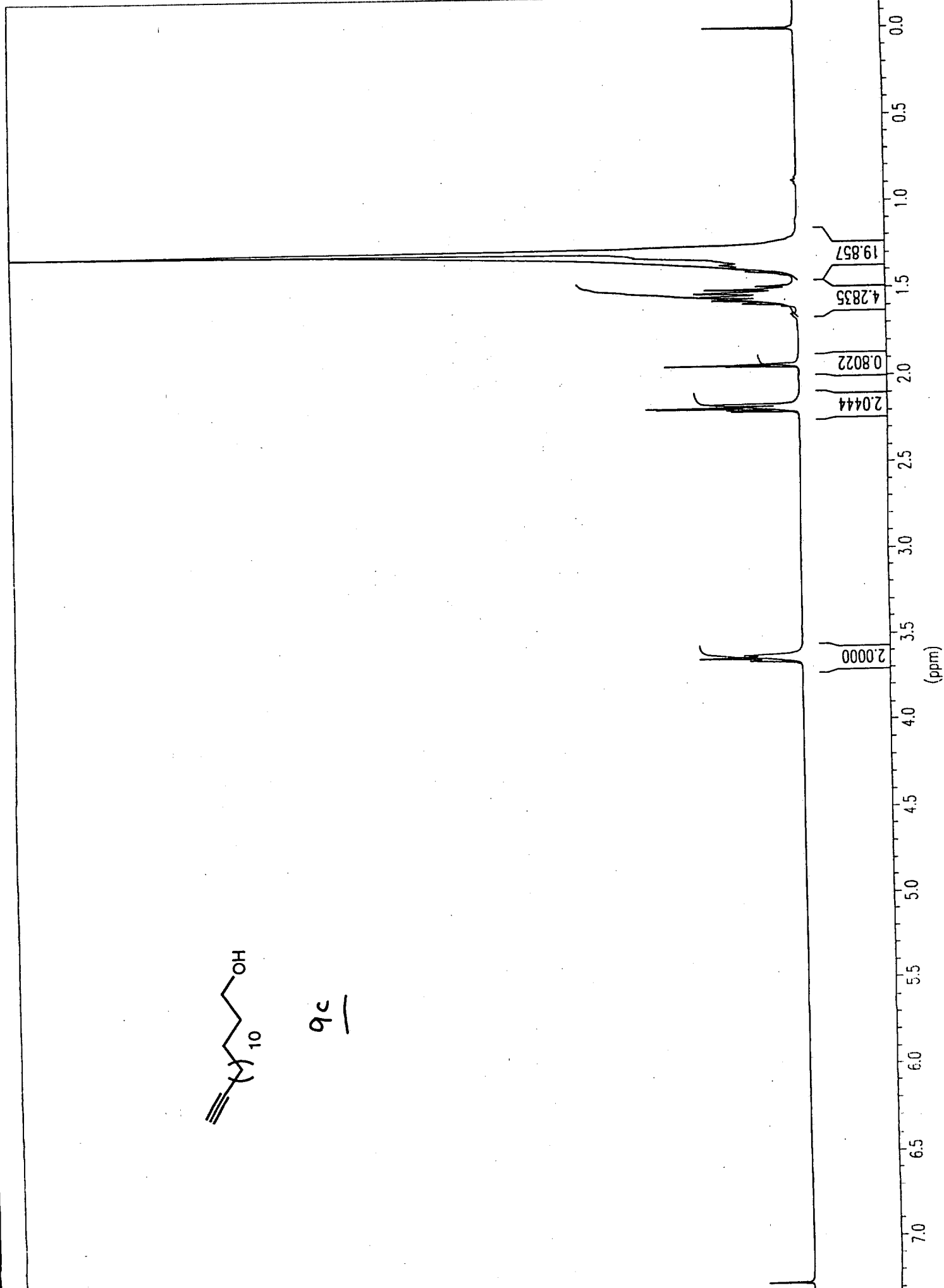


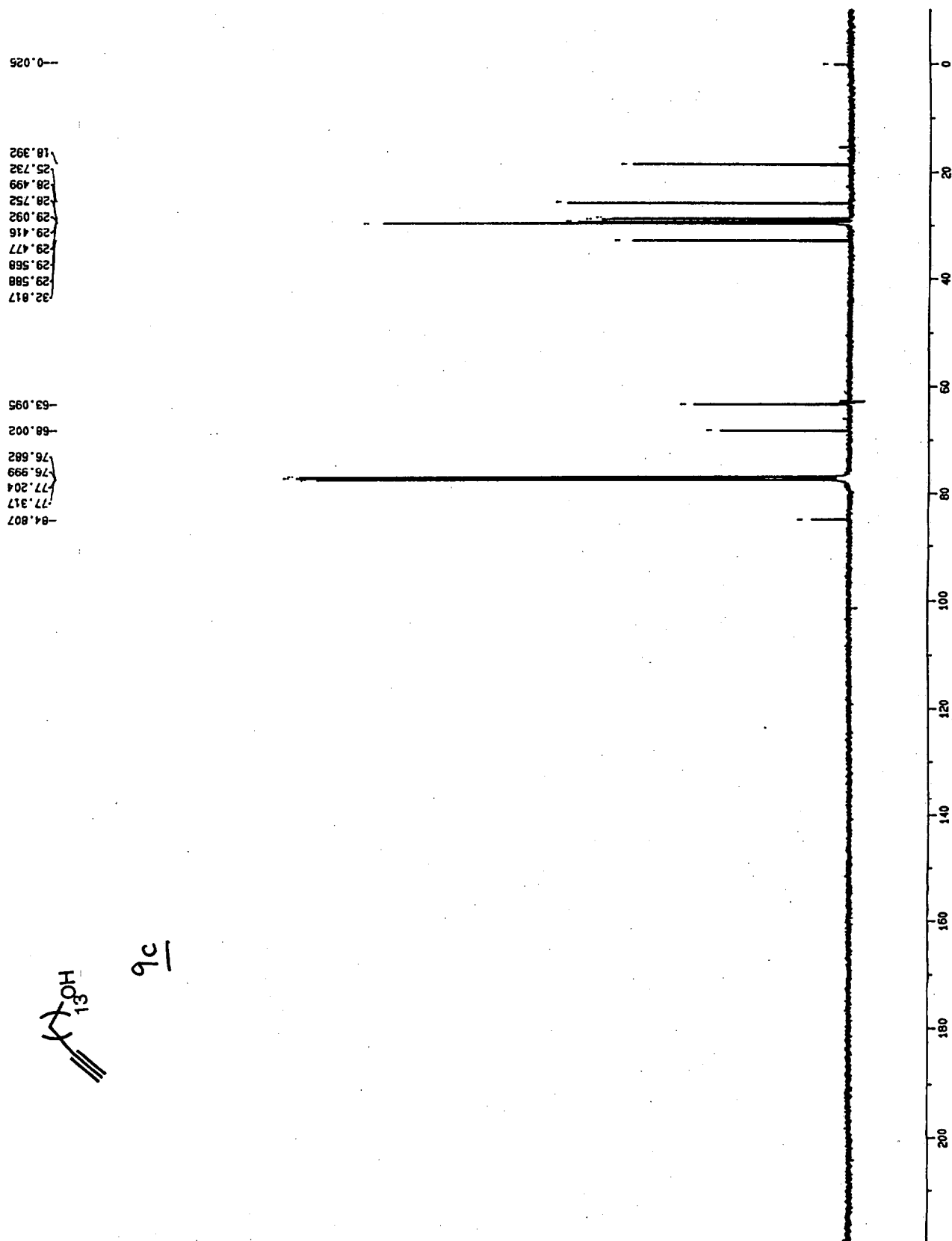
9a





9c





13C CPD PULS

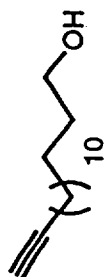
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 DE: 32768
 NUCLEUS: 13C
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 SH1: 25000.00 Hz
 TD: 65536
 NS: 8192
 DS: 2

F2 - Processing parameters
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1D NMR plot parameters
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 F1: 22638.12 Hz
 F2P: -10.000 ppa
 F2: -1006.14 Hz
 PPMCH: 7.26682 ppa/cm
 HZCH: 733.15527 Hz/cm

SPECTRUM : C, CH2 UP /CH, CH3 DOWN



9c

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76.3707

84.8004

68.0164

63.0564

29.5715
29.4808
29.0936
28.7518
28.4798
23.4100

32.7938

18.3822

25.7259

20 PPM

30

40

50

60

70

80

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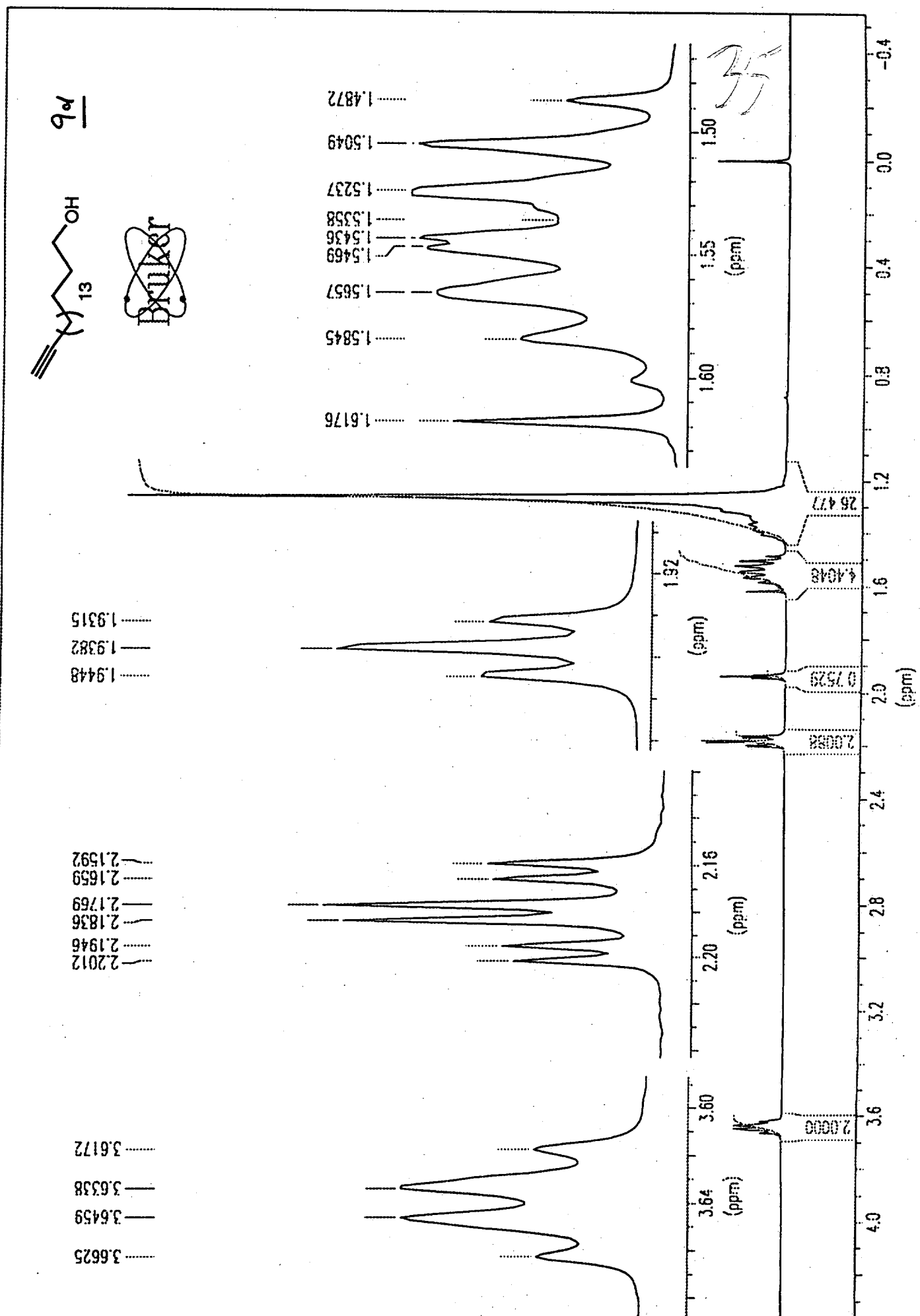
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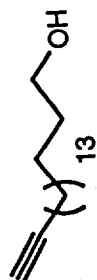
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 Reference

EXPERIMENT
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 Tube OD mm
 Temp °C
 Solvent CDCl3

SAMPLE
 EL SAVED ENAD FHC/030
 C13-APT 02-5.5 10-3 S
 JPS

Number
 File APT
 Date 02-23-95
 13.59.36
 XL GENA 200





9d

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75.681
-57.994
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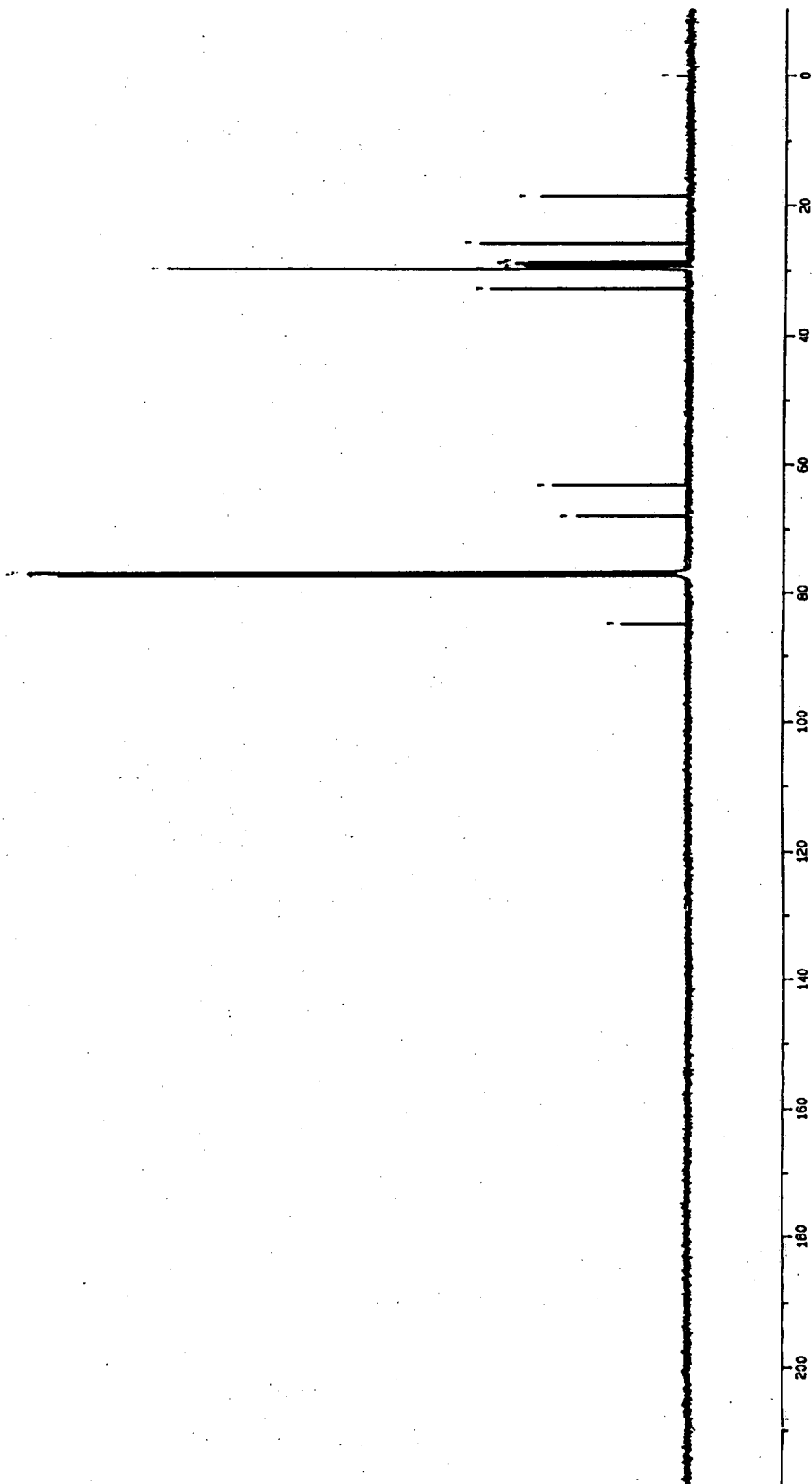
13C CPD PULS

Current Data Parameters
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EXPNO 20
PROCNO 1

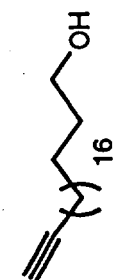
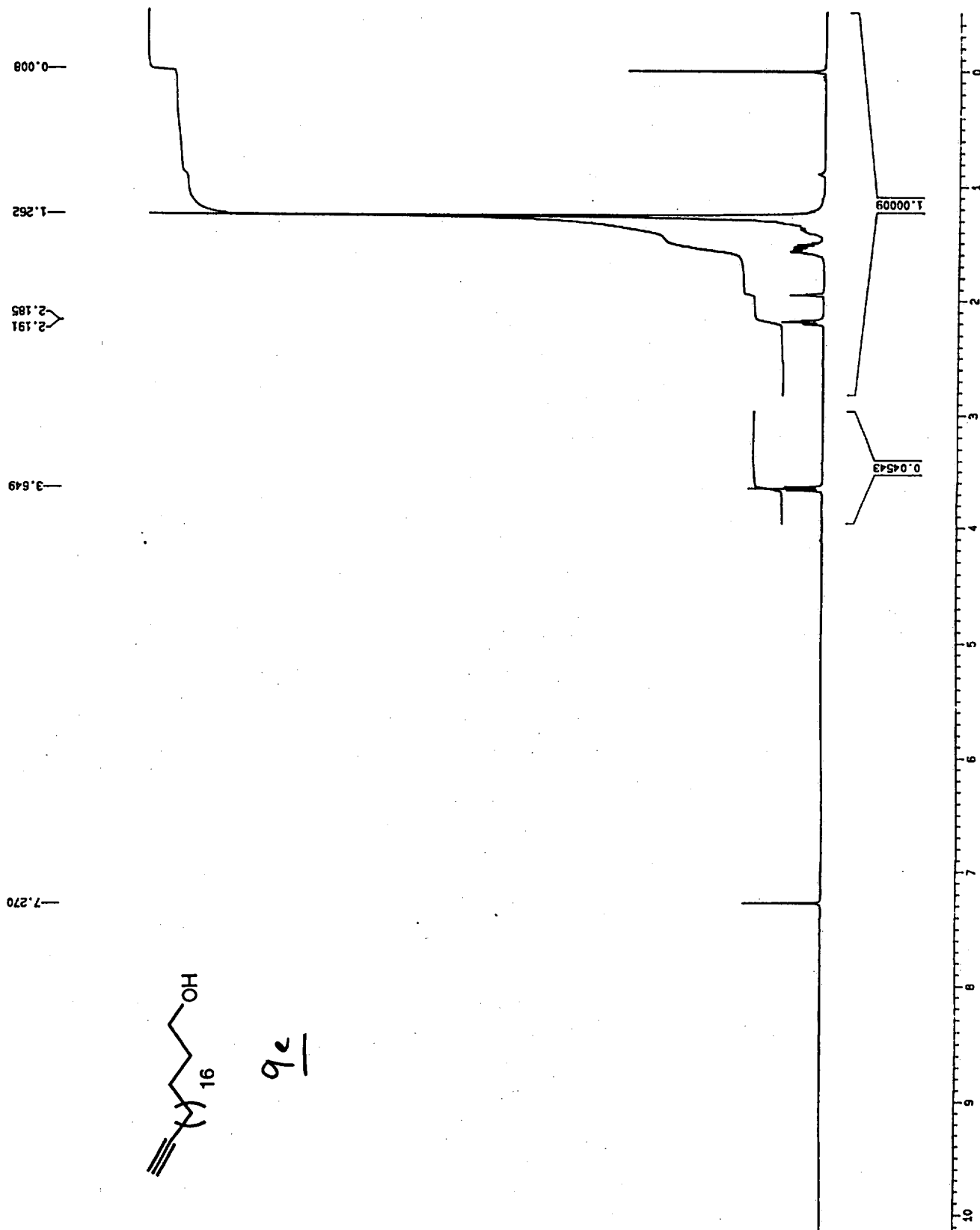
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D1 1.0000000 sec
S2 16 dB
P1 6.7 usec
DE 25.0 usec
SF01 100.6248040 MHz
SWH 25000.00 Hz
TD 65536
NS 1500
DS 2

F2 - Processing parameters
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SF 100.6136709 MHz
WDW EM
SSB 0
LB 1.00 Hz
GB 0
PC 1.00

1D NMR plot parameters
CX 32.25 cm
F1P 225.000 ppm
F1 22636.12 Hz
F2P -10.000 ppm
F2 -1006.14 Hz
PPMCH 7.28682 ppm/cm
HZCH 733.15327 Hz/cm



AMX 400
1H SPECTRUM



9c

Current Data Parameters
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EXPNO 10
PROCNO 1

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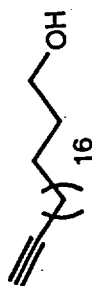
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1D NMR plot parameters
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F2P -0.500 ppm
F2 -200.07 Hz

13C CPD PULS

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28.489
28.742
29.078
29.405
29.471
29.574
29.643
32.808

63.071
67.967
76.657
76.975
77.292
84.774



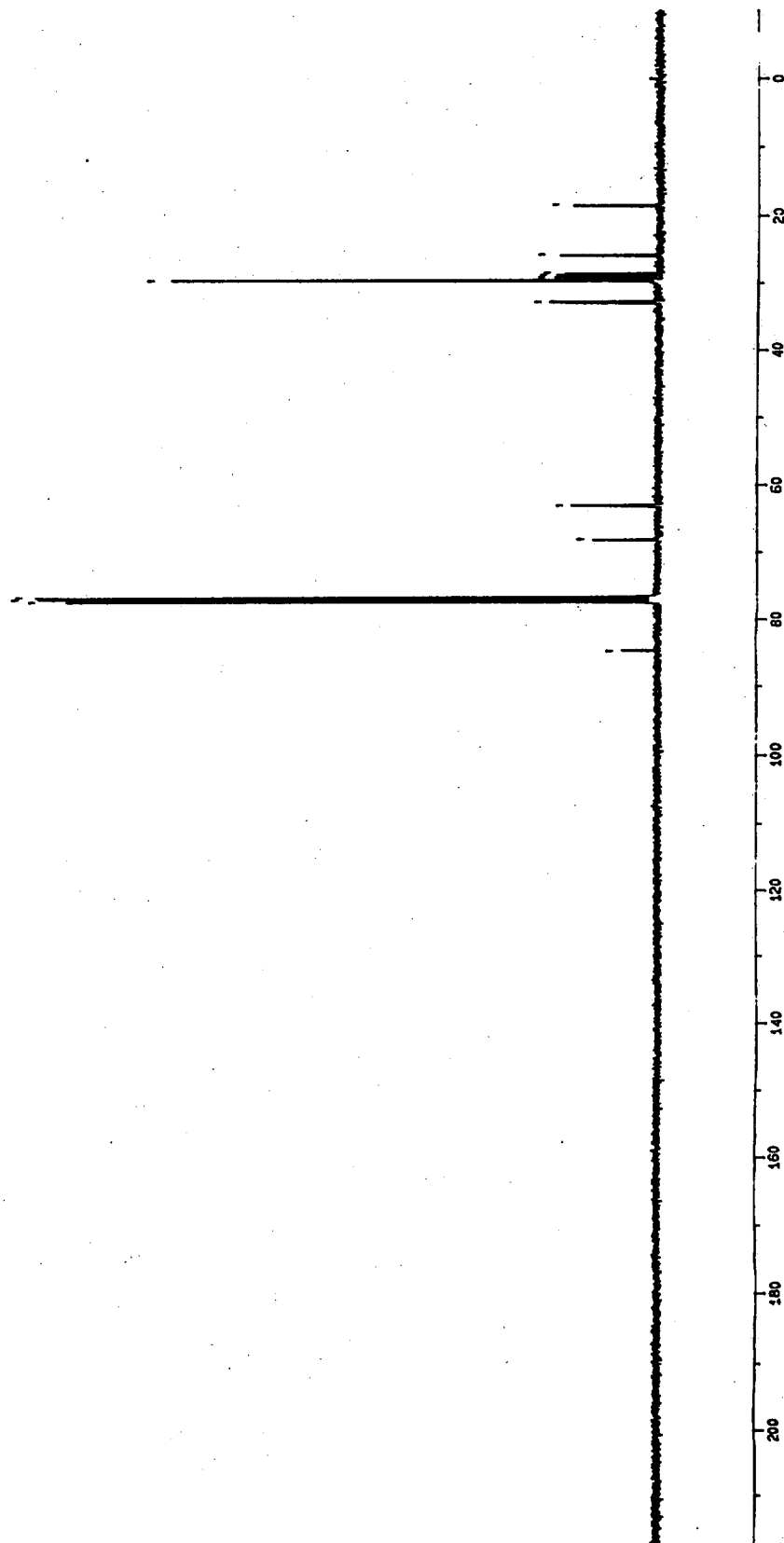
9e

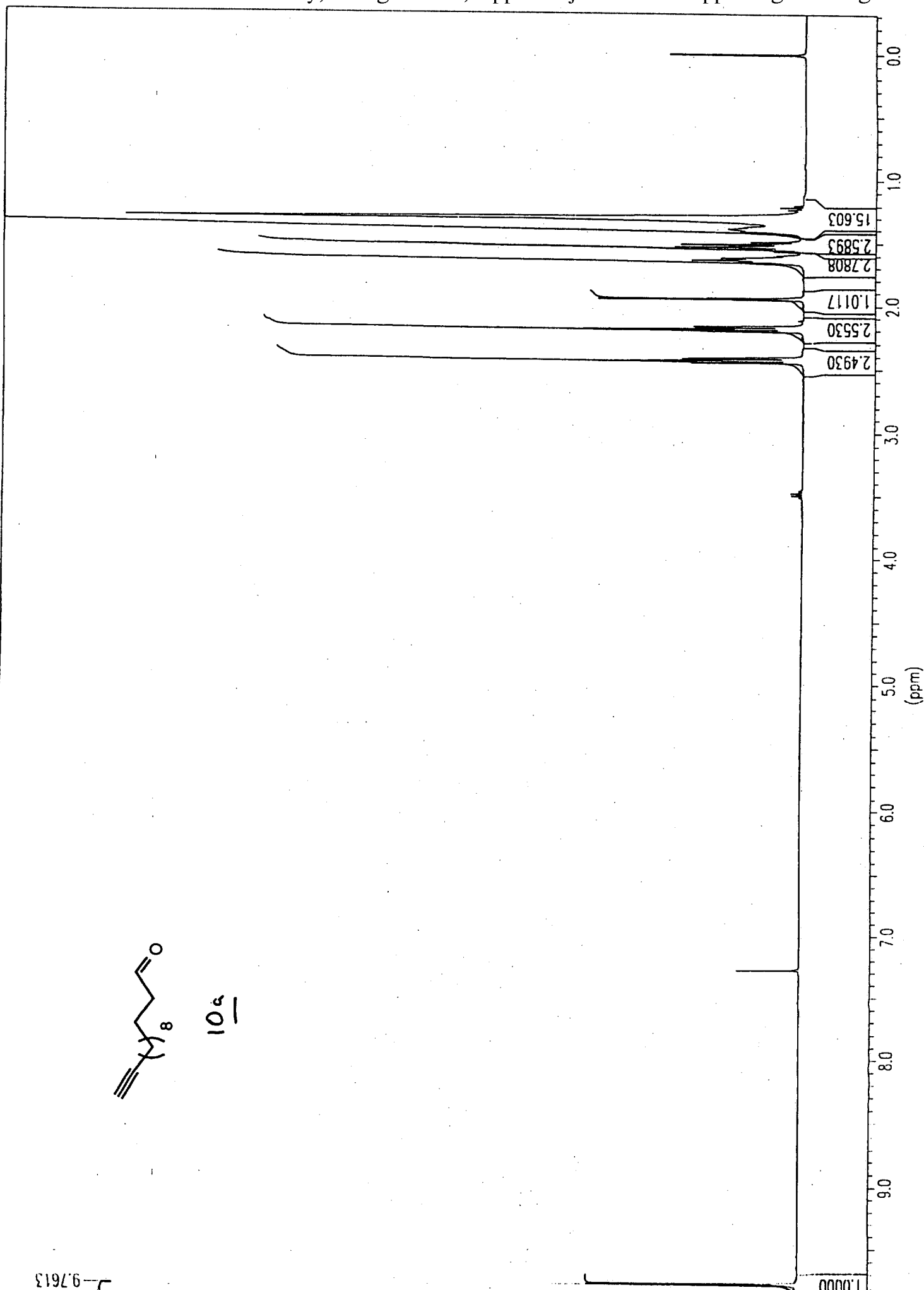
Current Data Parameters
NAME EME31
EXPNO 20
PROCNO 1

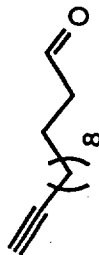
F2 - Acquisition Parameters
Date 981115
Time 4.46
PULPROG zg_cgpc30
SOLVENT CDCl3
AQ 1.3107400 sec
FIDRES 0.381470 Hz
DQ 20.0 usec
R6 32768
NUCLEUS 13C
H1 2 dB
D1 1.0000000 sec
S2 16 dB
P1 6.7 usec
DE 25.0 usec
SF01 100.6248040 MHz
SMH 25000.00 Hz
TD 65536
NS 1300
DS 2

F2 - Processing parameters
SI 32768
SF 100.6138716 MHz
WDW EM
SSB 0
LB 1.00 Hz
GB 0
PC 1.00

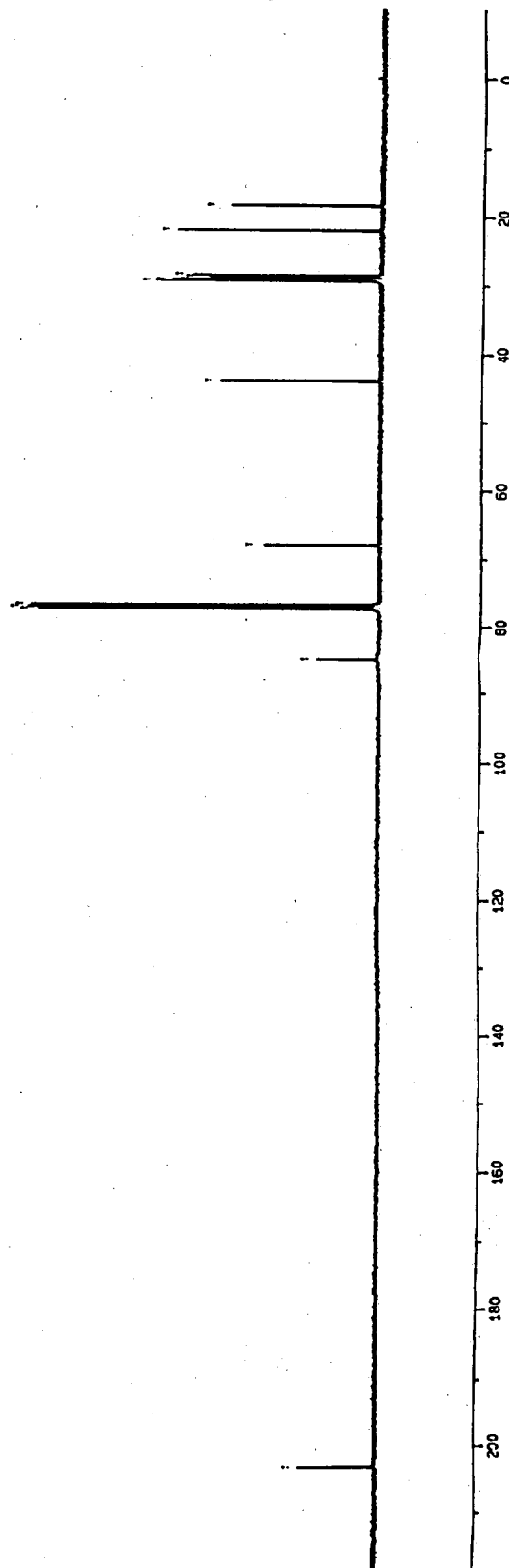
1D NMR plot parameters
CX 32.25 ca
F1P 225.000 ppm
F1 22638.12 Hz
F2P -10.000 ppm
F2 -1006.14 Hz







10a



13C CPD PULS

Current Data Parameters	
NAME	EX154
EXPNO	20
PROCNO	1

F2 - Acquisition Parameters	
Date	960927
Time	7.59
PULPROG	zg_cgdd30
SOLVENT	CDCl3
AQ	1.3107400 sec
FIDRES	0.381470 Hz
DM	20.0 usec
RG	32768
NUCLEUS	13C
HL1	2 dB
D1	1.0000000 sec
S2	18 dB
P2	6.7 usec
DE	25.0 usec
SF01	100.6248040 MHz
SWH	25000.00 Hz
TD	65536
NS	2048
DS	2

F2 - Processing parameters	
SI	32768
SF	100.6138731 MHz
WDW	EM
SSB	0
LB	1.00 Hz
GB	0
PC	1.00

1D NMR plot parameters	
CX	32.25 cm
F1	225.000 ppm
F2	22638.12 Hz
ZP	-10.000 ppm
F2	-1006.14 Hz
PPHCH	7.28682 ppm/cm
HZN	733.15533 Hz/cm

40